

Factor-Based Neural Network Forecasting of the German Yield Curve



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Abstract

This thesis develops a factor-based framework for forecasting the German zero-coupon yield curve, motivated by its role as the European Central Bank’s internal fair value model for euro area government bonds. Using daily data from 1999 to 2026, we compare Principal Component Analysis on yield changes and the Nelson-Siegel model as dimensionality reduction approaches, finding that PCA with four components provides superior and more stable reconstruction across all maturities and yield curve regimes. A key methodological contribution is the treatment of rolling PCA loadings as a locally stable coordinate system, ensuring that lagged factor inputs and forecast targets share a common basis across the full lag horizon.

We estimate a vector autoregression over the four principal component changes as the linear benchmark, establishing that the components are dynamically interdependent despite their cross-sectional orthogonality. An LSTM network is then evaluated across four variants with increasing information sets. We find that feeding the LSTM the coordinate-consistent factor representation, rather than raw factor differences, is essential for performance, and that the richest variant incorporating swaption implied volatility achieves the only positive out-of-sample R^2 for the fourth principal component across all specifications. At the yield curve level, no model beats the random walk on RMSE at the 21-trading-day horizon, consistent with the findings of Diebold and Li (2006). However, a hybrid model combining VAR predictions for the level and curvature factors with LSTM predictions for the slope and higher-order factors achieves the strongest directional accuracy at longer tenors, suggesting that nonlinear dynamics in the higher-order seem present.

Contents

1	Introduction	5
2	Literature Review	7
2.1	Dimensionality Reduction	7
2.1.1	Parametric Dimensionality Reduction: The Nelson-Siegel Model and Its Extensions	7
2.1.2	Statistical Dimensionality Reduction: Principal Component Analysis	7
2.2	Dynamic Factor Forecasting: The Diebold-Li Framework	8
2.3	Neural Networks for Yield Curve Analysis	8
2.3.1	Long Short-Term Memory: Hochreiter and Schmidhuber (1997) and Gers (1999)	9
2.3.2	LSTM-Based Yield Curve Forecasting: Suimon et al. (2020)	9
2.3.3	Neural Network Forecasting with Market Variables: Verner et al. (2021)	10
3	Data	11
3.1	Yield Curve Data	11
3.2	Swaption Implied Volatility Data	13
4	Yield Curve Fitting	14
4.1	Motivation	14
4.2	Nelson-Siegel	14
4.3	Principal Component Analysis on Yield Changes	15
4.4	A Consistent Factor Coordinate System	15
4.5	Loading Stability	16
4.6	Reconstruction Assessment	17
5	Forecasting Principal Components with VAR	20
5.1	Autocorrelation Analysis and Lag Selection	20
5.2	VAR Specification and In-Sample Results	22
5.3	Rolling VAR and Out-of-Sample Results	23
5.3.1	Training Window Selection	23
5.3.2	Out-of-Sample Results	25
6	Forecasting Principal Components with LSTM	26
6.1	Feature Construction	26
6.1.1	Sequential Features	26
6.1.2	Static Features	26
6.1.3	Feature Standardisation	26
6.2	Model Variants	27
6.3	Architecture and Training	28
6.3.1	LSTM Layers	28
6.3.2	Feedforward Head	28
6.3.3	Output Layer and Scaled Tanh	28
6.3.4	Training Procedure	29
6.4	Out-of-Sample Results	29
6.4.1	LSTM-VAR versus VAR	29
6.4.2	LSTM-FULL versus LSTM-VAR and VAR	30
6.4.3	LSTM-FULL-EIG versus LSTM-FULL	30

6.4.4	Combined Variant	30
7	Yield Curve Reconstruction	31
7.1	From Factor Forecasts to Yield Levels	31
7.2	Results	31
7.2.1	Random Walk Benchmark	31
7.2.2	VAR versus LSTM	32
7.2.3	Hybrid Model	32
8	Conclusion	32
	Appendix	34
A1	Loading Stability Results	34
A2	Reconstruction Error by Tenor	36
A3	ACF and PACF Results	38
A4	In-Sample VAR Results	39
A5	Out-of-Sample LSTM Forecast Results	44
A6	Yield Curve Reconstruction Residual Statistics	46

1 Introduction

The yield curve, describing the relationship between interest rates and time to maturity for government bonds, plays a central role in monetary policy and fixed-income markets. For the European Central Bank, with whom this research is conducted, the German zero-coupon yield curve serves as an internal fair value model: constructed daily from all outstanding German government bonds, it provides the benchmark against which individual securities are assessed as cheap or expensive relative to the market. Anticipating which parts of the term structure are likely to shift over the coming month gives direct insight into where mispricing may emerge and supports forward-looking decisions on asset purchases and interest rate risk management.

This thesis develops a factor-based forecasting framework for that curve. We first establish that PCA on yield changes dominates Nelson-Siegel in reconstruction accuracy and adopt it as the basis for forecasting. We then formalise the treatment of rolling PCA loadings as a stable coordinate system, ensuring that all model inputs are expressed in a common basis. We estimate a vector autoregression over the resulting factor series as the linear benchmark, and evaluate whether an LSTM network can capture nonlinear dynamics beyond it. Finally, we investigate whether swaption implied volatility carries incremental predictive information for factor movements.

Dimensionality reduction is the natural starting point for yield curve forecasting. Litterman and Scheinkman (1991) establish empirically that three principal components account for nearly all yield curve variation, corresponding to level, slope, and curvature. Nelson and Siegel (1987) propose a parametric alternative that represents the same structure with three factors governed by a single decay parameter λ . Both approaches motivate the same two-stage procedure used by Diebold and Li (2006): extract a small number of factors, then forecast them as a time series. The key difference is that PCA imposes no assumption on the shape of factor loadings and produces orthogonal factors by construction, which avoids the estimation instability documented for four-factor parametric extensions Gilli et al. (2010). We evaluate both methods on German zero-coupon yield data and find PCA on yield changes to be the superior representation.

A contribution of this thesis is the treatment of rolling PCA loadings as a locally stable coordinate system. When lagged factor values are used as model inputs, each lag is typically projected onto its own loading matrix, making the resulting coordinates incomparable across time. We demonstrate empirically that the loading matrix $\mathbf{L}_t$ changes negligibly over a one-month horizon, which allows all lagged observations to be projected onto the same $\mathbf{L}_t$. This ensures that inputs and forecast targets share a common basis, a requirement for a well-posed sequential learning problem.

To exploit the autocorrelation structure in the factor series, we estimate a vector autoregression following Diebold and Li (2006). A non-obvious finding is that the four principal components, while orthogonal by construction, are dynamically interdependent: lagged values of the level factor are significant predictors in the equations for all other components, making the joint VAR the appropriate specification over univariate alternatives.

To capture potential nonlinear dynamics beyond the VAR, we employ a Long Short-Term Memory network Hochreiter and Schmidhuber (1997); Gers (1999). LSTMs are well-suited

to this setting because the coordinate-consistent representation described above provides a temporally structured input across eleven lag positions, and the LSTM’s selective memory can in principle exploit dependencies that the linear VAR misses. We evaluate several variants with increasing information sets and find that the coordinate-consistent representation, rather than the architecture itself, is the key driver of performance.

As a fourth component, we investigate whether $3M \times 10Y$ ATM normal swaption implied volatility carries incremental predictive information beyond the factor history. Implied volatility reflects market participants’ uncertainty about future rate dynamics and may therefore signal regime changes that are not yet visible in realised factor movements. Motivated by Verner et al. (2021), who examine external market variables in fixed-income neural network forecasting, we include it as a sequential feature in the richest LSTM variant and find that it contributes to the only positive OOS R^2 observed for the fourth principal component.

The remainder of this thesis is organised as follows. Section 2 reviews the relevant literature. Section 3 describes the data. Section 4 compares the dimensionality reduction approaches and validates the loading stability assumption. Section 5 presents the VAR forecasting results. Section 6 develops the LSTM framework and evaluates its out-of-sample performance against the VAR. Section 7 reconstructs yield curve forecasts from the component-level predictions and evaluates them against a random walk benchmark. Section 8 concludes.

2 Literature Review

The literature on yield curve modeling and forecasting has evolved considerably over the past decades, progressing from parametric curve-fitting methods to factor-based forecasting frameworks, and more recently incorporating machine learning techniques. This review traces this evolution, beginning with foundational work on dimensionality reduction, proceeding through the development of dynamic factor forecasting models, and concluding with recent applications of neural networks to yield curve analysis. This review is structured to highlight how each contribution addresses specific aspects of the yield curve forecasting problem.

2.1 Dimensionality Reduction

Forecasting the yield curve presents an inherent high-dimensionality challenge: yields across many maturities must be predicted simultaneously, yet modelling each maturity independently ignores the strong co-movement that characterises term structure dynamics. A small number of common factors explains the vast majority of yield variation Litterman and Scheinkman (1991), making dimensionality reduction a natural first step that transforms the forecasting problem from predicting many correlated yields to predicting a handful of interpretable factors.

2.1.1 Parametric Dimensionality Reduction: The Nelson-Siegel Model and Its Extensions

The challenge of modeling yield curves across numerous maturities was first addressed systematically by Nelson and Siegel (1987), who proposed a parsimonious functional form capable of generating the diverse shapes observed in practice. The Nelson-Siegel model represents the yield curve at time t for maturity τ as:

$$y_t(\tau) = \beta_{1,t} + \beta_{2,t} \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} \right) + \beta_{3,t} \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau} \right) \quad (1)$$

where $\beta_{1,t}$, $\beta_{2,t}$, and $\beta_{3,t}$ are time-varying parameters representing level, slope, and curvature respectively, and λ is a fixed decay parameter. By reducing an N -dimensional yield curve to just three parameters, Nelson and Siegel (1987) demonstrated that complex yield curve shapes could be parsimoniously represented while maintaining economic interpretability.

Svensson (1994) extended this framework by introducing a fourth factor with an additional curvature term, increasing flexibility in fitting irregular curve shapes. This Nelson-Siegel-Svensson (NSS) model has become standard practice among central banks BIS (2005). However, the two curvature factors are prone to confounding effects, where the parameters interact and can take on extreme values to offset one another, producing unstable estimates Gilli et al. (2010). This instability motivates the use of the three-factor Nelson-Siegel specification in this thesis rather than its four-factor extension, as discussed further in Section 4.

2.1.2 Statistical Dimensionality Reduction: Principal Component Analysis

Litterman and Scheinkman (1991) took a different approach by applying principal component analysis (PCA) to U.S. Treasury yields, letting the data reveal the dominant factors driving yield curve variation rather than imposing a parametric structure. The PCA approach offers two advantages over parametric methods. First, it requires no assumption about the

functional form of factor loadings. Second, it provides a natural measure of how much variance each factor explains, allowing researchers to determine the appropriate number of factors empirically. Crucially, unlike the NSS model, PCA factors are orthogonal by construction, which prevents the parameter explosion documented in the parametric setting Gilli et al. (2010). This makes it natural to retain a fourth principal component without the instability concerns associated with the NSS fourth factor, a choice we revisit and justify empirically in Section 4.

2.2 Dynamic Factor Forecasting: The Diebold-Li Framework

While Nelson and Siegel (1987) and Litterman and Scheinkman (1991) established that yield curves can be effectively represented by three factors, Diebold and Li (2006) made the crucial step of creating a forecasting framework by treating these factors as time series to be predicted. Diebold and Li (2006) combined the Nelson and Siegel (1987) factor structure with vector autoregression (VAR) models to forecast the entire yield curve. Their two-stage approach works as follows. First, at each point in time, they estimate the three Nelson and Siegel (1987) factors $\beta_{1,t}$, $\beta_{2,t}$, and $\beta_{3,t}$ by fitting equation (1) to observed yields. Second, they model the dynamics of these factors using a VAR(1) specification:

$$\begin{bmatrix} \beta_{1,t} \\ \beta_{2,t} \\ \beta_{3,t} \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} + \begin{bmatrix} \phi_{11} & \phi_{12} & \phi_{13} \\ \phi_{21} & \phi_{22} & \phi_{23} \\ \phi_{31} & \phi_{32} & \phi_{33} \end{bmatrix} \begin{bmatrix} \beta_{1,t-1} \\ \beta_{2,t-1} \\ \beta_{3,t-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \\ \varepsilon_{3,t} \end{bmatrix} \quad (2)$$

To generate forecasts h periods ahead, they iterate the VAR forward and then reconstruct the yield curve at each horizon using the Nelson and Siegel (1987) formula 1 with the forecasted factor values. This approach transforms a high-dimensional forecasting problem (predicting yields at many maturities) into a low-dimensional one (predicting three factors).

The key empirical finding of Diebold and Li (2006) is that this factor-based VAR approach significantly outperforms a simple random walk benchmark in out-of-sample forecasts, particularly at medium to long horizons. The random walk model, had proven difficult to beat in earlier research, making this result notable. The success of the Diebold and Li (2006) framework stems from exploiting the cross-sectional restrictions implied by the factor structure: rather than forecasting each maturity independently, the model recognizes that yields across maturities move together in predictable ways captured by the three factors. This framework has become the benchmark against which subsequent yield curve forecasting methods are evaluated.

However, the Diebold and Li (2006) approach has an important limitation: the VAR specification assumes that the relationship between past and future factor values is linear. While this assumption simplifies estimation and interpretation, it may be restrictive if yield curve dynamics exhibit regime changes, threshold effects, or other nonlinearities. Moreover, the model relies exclusively on the historical behavior of the factors themselves, ignoring potentially valuable information from other market variables that might help predict future movements.

2.3 Neural Networks for Yield Curve Analysis

The application of neural networks to yield curve modeling has taken several distinct forms, addressing different aspects of the curve analysis problem. We review three key contributions

that collectively motivate this approach: one introducing the LSTM architecture, one applying LSTM directly to yield curve forecasting, and one incorporating external market variables into neural network-based fixed-income forecasting.

2.3.1 Long Short-Term Memory: Hochreiter and Schmidhuber (1997) and Gers (1999)

Hochreiter and Schmidhuber (1997) introduced the Long Short-Term Memory (LSTM) network to address the vanishing gradient problem that prevented standard recurrent neural networks from learning dependencies across long time intervals. The key innovation is the memory cell, which maintains its state across timesteps through a constant error carousel, with multiplicative input and output gates controlling access to the stored information.

The original architecture was subsequently extended by Gers (1999), who identified a weakness in processing continuous input streams: without a mechanism to reset cell state, the memory can grow indefinitely and cause network saturation. Their remedy is the forget gate, an adaptive mechanism that learns when to clear the cell state, completing the three-gate architecture that is standard in modern implementations including TensorFlow. This thesis employs this complete architecture, and both papers are therefore cited as the foundation of the LSTM component.

2.3.2 LSTM-Based Yield Curve Forecasting: Suimon et al. (2020)

Beyond the autoencoder for dimensionality reduction, Suimon et al. (2020) implement a Long Short-Term Memory (LSTM) model to directly forecast the yield curve one month ahead using historical interest rate data. This forms the more relevant methodological comparison for the forecasting component of yield curve modeling.

The LSTM model takes as input the full yield curve across maturities (2-, 5-, 7-, 10-, 15-, and 20-year rates) at four consecutive weekly timesteps. Three weeks prior, two weeks prior, one week prior, and the current date are used. This sequence is mapped to the predicted yield curve one month later:

$$\hat{\mathbf{y}}_{t+h} = \text{LSTM}(\mathbf{y}_t, \mathbf{y}_{t-1}, \mathbf{y}_{t-2}, \mathbf{y}_{t-3}) \quad (3)$$

where $\mathbf{y}_{t-k}$ denotes the vector of observed yields k weeks before the investment date and h represents the one-month forecast horizon. Two LSTM variants are tested: LSTM20 and LSTM50, with 20 and 50 units respectively, both using hyperbolic tangent activation and sigmoid recurrent activation, consistent with the standard LSTM architecture of Hochreiter and Schmidhuber (1997). The model is trained on a rolling window of 5 years of weekly data and retrained annually.

Investment decisions are based on the direction of predicted yield changes: if the model forecasts a yield increase at a given maturity, the corresponding government bond is shorted; if a decrease is forecast, the bond is bought. The investment horizon is one month.

The LSTM outperforms both the autoencoder strategy and the trend-following benchmark in terms of cumulative capital gains, particularly across the 10- and 20-year maturity government bonds. Suimon et al. (2020) attribute this advantage to the LSTM's ability

to exploit temporal dependencies in historical interest rate data, which the autoencoder that operates only on the current curve shape cannot capture. The VAR model performs comparably to the LSTM, suggesting that much of the predictive signal is captured already by linear dynamics, with the LSTM providing additional but modest gains through nonlinear pattern recognition.

A limitation noted by Suimon et al. (2020) is the reduced interpretability of the LSTM strategy relative to the autoencoder approach: it is not straightforward to attribute the LSTM's trading signal to either trend-following behaviour or curve-shape mean reversion, as both mechanisms may simultaneously drive the predictions.

2.3.3 Neural Network Forecasting with Market Variables: Verner et al. (2021)

Verner et al. (2021) applied nonlinear autoregressive neural networks (NAR and NARX architectures) to forecast long-term bond prices, representing one of the first attempts to incorporate external market variables into neural network-based fixed-income forecasting. Their approach involved forecasting individual bond prices (rather than yield curves) using two types of models: a pure autoregressive network using only past bond prices, and an augmented network incorporating 50-year EUR interest rate swaps and the VIX volatility index as external inputs.

The NARX (Nonlinear Autoregressive with eXogenous inputs) architecture they employed can be represented as:

$$P_t = f(P_{t-1}, P_{t-2}, \dots, P_{t-d}, X_t, X_{t-1}, \dots, X_{t-d}) + \varepsilon_t \quad (4)$$

where P_t is the bond price, X_t represents external variables (swap rates and VIX), and $f(\cdot)$ is approximated by a feedforward neural network with multiple hidden layers. Their results showed that both the Levenberg-Marquardt and Scaled Conjugate Gradient training algorithms achieved high in-sample fit ($R^2 > 95\%$), demonstrating the capability of neural networks to model bond price dynamics.

However, their key finding regarding external variables was somewhat negative: incorporating VIX and swap rates provided only marginal improvement over models using price history alone. This result has several potential explanations. First, their focus on individual bond prices rather than the entire yield curve structure means they did not exploit the cross-sectional information that factor models capture. Second, VIX represents equity market volatility and may not be the most relevant measure for fixed-income forecasting; curve-specific measures like swaption implied volatility might be more informative.

3 Data

3.1 Yield Curve Data

The zero-coupon yield curve data for Germany are obtained from an internal ECB database. The curve is constructed by collecting all outstanding German government bonds at each point in time from the IBoxx bond database, a product of LSEG (formerly Markit), computing zero-coupon yields from observed bond prices, and fitting a locally weighted scatterplot smoothing (LOWESS) spline Cleveland (1979) across maturities to obtain a continuous term structure. This curve constitutes the ECB's fair value model of the German term structure and serves as the basis for ECB asset purchase decisions, making it the most relevant benchmark for euro area interest rate dynamics.

The dataset spans 4 January 1999 to 20 April 2026, comprising 6,986 daily observations after excluding weekends. Yields are available at 15 maturity points: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 15, 20, 25, and 30 years. No missing observations are present in the yield curve series.

Table 1 presents summary statistics across all maturities. Where Figure 1 illustrates the full evolution of the yield curve over the sample period.

Table 1: Summary statistics of German zero-coupon yields of ECB fair value model, 1999–2026. $N = 6,986$ daily observations.

Maturity	Mean	Std. Dev.	Min	Q25	Median	Q75	Max
1Y	0.015	0.018	−0.009	−0.002	0.015	0.030	0.051
2Y	0.016	0.018	−0.010	−0.002	0.017	0.030	0.052
3Y	0.017	0.018	−0.010	−0.001	0.019	0.030	0.052
4Y	0.018	0.018	−0.010	−0.001	0.021	0.033	0.052
5Y	0.019	0.018	−0.010	0.000	0.022	0.035	0.053
6Y	0.020	0.018	−0.010	0.002	0.023	0.036	0.053
7Y	0.021	0.018	−0.010	0.003	0.023	0.037	0.055
8Y	0.023	0.018	−0.009	0.004	0.024	0.039	0.055
9Y	0.023	0.018	−0.009	0.005	0.025	0.039	0.056
10Y	0.024	0.018	−0.009	0.007	0.026	0.040	0.056
12Y	0.026	0.017	−0.008	0.008	0.027	0.041	0.057
15Y	0.027	0.017	−0.008	0.010	0.028	0.042	0.058
20Y	0.029	0.018	−0.007	0.012	0.030	0.045	0.059
25Y	0.031	0.019	−0.006	0.013	0.030	0.047	0.068
30Y	0.030	0.018	−0.005	0.014	0.030	0.045	0.063

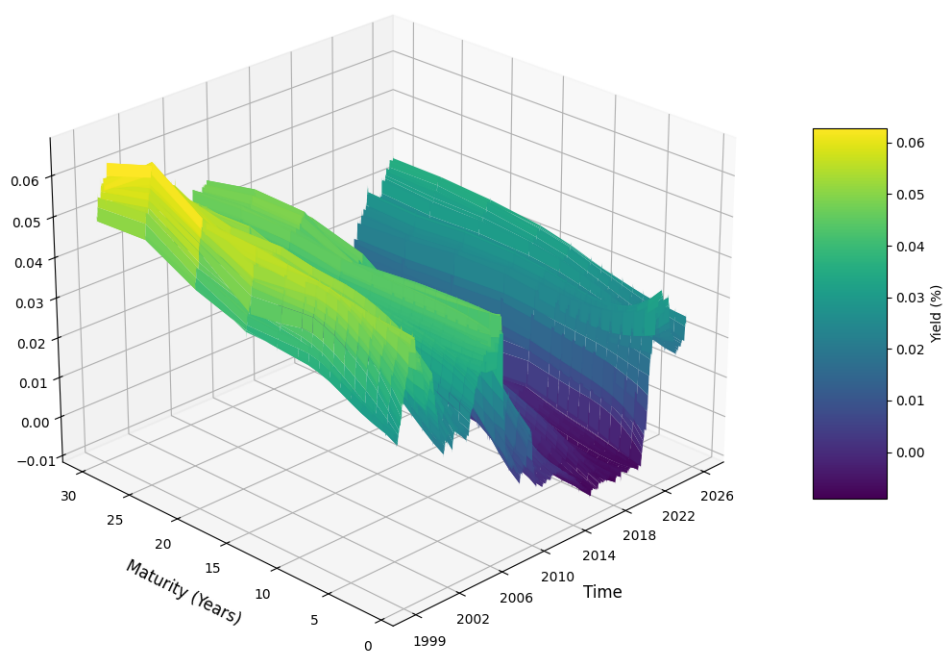


Figure 1: Evolution of the German zero-coupon yield curve, 1999–2026.

3.2 Swaption Implied Volatility Data

The implied volatility series is the Merrill Lynch Swaption Option Volatility Estimate with a 3-month option tenor on a 10-year underlying swap ($3M \times 10Y$), sourced from Bloomberg Finance L.P. and obtained via an internal ECB database. The volatility is quoted as at-the-money (ATM) normal (basis-point) implied volatility, expressed in basis points per annum.

The series spans the same sample period as the yield curve data. For 96 dates where observations are missing, values are recovered by linear interpolation between the nearest available observations before and after the gap:

$$IV_t = IV_{t-k} + \frac{t - (t-k)}{(t+j) - (t-k)} (IV_{t+j} - IV_{t-k}) \quad (5)$$

where k and j denote the number of days to the nearest observed values before and after t . All gaps are short, consisting of isolated missing dates, so interpolation introduces negligible distortion.

Table 2 presents summary statistics. Where Figure 2 shows the full time series.

Table 2: Summary statistics of $3M \times 10Y$ ATM normal swaption implied volatility, 1999–2026. Values in basis points. $N = 6,986$.

	Mean	Std. Dev.	Min	Q25	Median	Q75	Max
IV (bp)	66.7	25.1	26.3	49.6	62.5	79.5	180.3

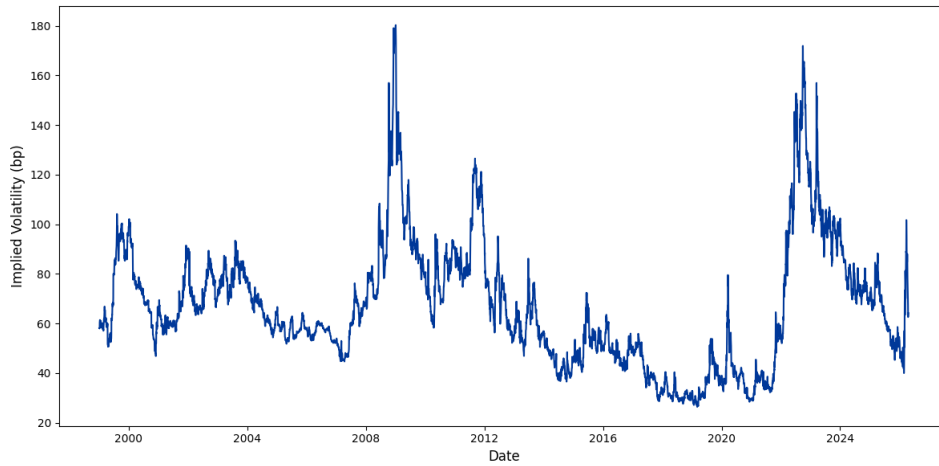


Figure 2: $3M \times 10Y$ ATM normal swaption implied volatility, 1999–2026. Values in basis points.

4 Yield Curve Fitting

4.1 Motivation

Forecasting the term structure of interest rates presents a high-dimensional challenge: modelling 15 maturities simultaneously requires capturing complex cross-maturity dependencies. This motivates dimensionality reduction: a small number of factors can explain most of the variation in the yield curve. Litterman and Scheinkman (1991) establish this empirically, showing that three principal components account for nearly all yield curve variation and correspond interpretably to level, slope, and curvature. Nelson and Siegel (1987) propose a parametric alternative, representing the yield curve with three factors governed by a single decay parameter λ . Diebold and Li (2006) demonstrate that such low-dimensional representations form an effective basis for yield curve forecasting.

We evaluate two approaches: Principal Component Analysis on yield changes and the Nelson-Siegel model. PCA is applied to yield *changes* rather than levels, as stationarity of the input series is required for the sample covariance matrix to converge.

Motivated by the four-factor extension of Svensson (1994), we test both three and four principal components. We do not consider the four-beta Nelson-Siegel- Svensson model, as Gilli et al. (2010) document severe optimisation instability in the NSS parameterisation: the additional decay parameter renders the loading matrix ill-conditioned, making reliable rolling estimation without additional constraints infeasible. PCA's orthogonality constraint avoids this issue, allowing a fourth factor to be retained without instability. The following subsections develop the mathematical framework for each method and evaluate their empirical fit on German zero-coupon yield data.

4.2 Nelson-Siegel

The Nelson-Siegel model Nelson and Siegel (1987) represents the yield curve parametrically as:

$$y(M) = \beta_0 + \beta_1 \frac{1 - e^{-\lambda M}}{\lambda M} + \beta_2 \left(\frac{1 - e^{-\lambda M}}{\lambda M} - e^{-\lambda M} \right) \quad (6)$$

where M denotes maturity and $\lambda > 0$ controls the decay rate of the loadings. In matrix form, for $m = 15$ maturities:

$$\mathbf{y}_t = \mathbf{L}(\lambda_t) \boldsymbol{\beta}_t \quad (7)$$

where $\mathbf{y}_t \in \mathbb{R}^m$, $\mathbf{L}(\lambda_t) \in \mathbb{R}^{m \times 3}$ contains the maturity-dependent loadings fully determined by λ_t , and $\boldsymbol{\beta}_t = [\beta_0, \beta_1, \beta_2]^T$ are the factor weights estimated by OLS given λ_t . The optimal λ_t is estimated by minimising reconstruction error over a rolling window of $\tau = 756$ trading days.

Forecasting $h = 21$ days ahead requires that the loading matrix remains approximately stable over the forecast horizon:

$$\lambda_{t+h} \approx \lambda_t \quad (8)$$

Since $\mathbf{L}(\lambda_t)$ is fully determined by λ_t , this is equivalent to $\mathbf{L}(\lambda_{t+h}) \approx \mathbf{L}(\lambda_t)$. Given a stable λ_t , future factor weights $\boldsymbol{\beta}_{t+h}$ are refitted by OLS at $t+h$ and the yield curve is reconstructed as:

$$\hat{\mathbf{y}}_{t+h} = \mathbf{L}(\lambda_t) \boldsymbol{\beta}_{t+h}(\lambda_t) \quad (9)$$

Assumption (8) is tested empirically in Section 4.5.

4.3 Principal Component Analysis on Yield Changes

PCA is applied to yield changes rather than levels, as stationarity of the input series is required for the sample covariance matrix to converge properly. The covariance matrix is estimated from daily yield changes $\delta \mathbf{y}_t = \mathbf{y}_t - \mathbf{y}_{t-1} \in \mathbb{R}^m$ over a rolling window of $\tau = 756$ trading days:

$$\hat{\Sigma}_t^\delta = \frac{1}{\tau - 1} \sum_{s=t-\tau}^{t-1} (\delta \mathbf{y}_s - \overline{\delta \mathbf{y}_t})(\delta \mathbf{y}_s - \overline{\delta \mathbf{y}_t})^T \quad (10)$$

Rather than fitting on overlapping 21-day changes, where consecutive observations share 20 out of 21 days and introduce artificial serial correlation, we fit on daily changes and scale under the i.i.d. assumption:

$$\hat{\Sigma}_t^\Delta = 21 \cdot \hat{\Sigma}_t^\delta \quad (11)$$

yielding 756 approximately independent observations per estimate. The eigendecomposition of $\hat{\Sigma}_t^\Delta$ gives:

$$\hat{\Sigma}_t^\Delta = \mathbf{V}_t \mathbf{\Lambda}_t \mathbf{V}_t^T \quad (12)$$

where $\mathbf{\Lambda}_t = \text{diag}(\nu_{1,t}, \dots, \nu_{m,t})$ contains eigenvalues in descending order. Since multiplying by a scalar does not alter eigenvectors, $\mathbf{V}_t$ is identical whether computed from daily or 21-day changes. The loading matrix retains the k leading eigenvectors:

$$\mathbf{L}_t = \mathbf{V}_t[:, 1 : k] \in \mathbb{R}^{m \times k} \quad (13)$$

where $k \in \{3, 4\}$. Factors are extracted by projecting the 21-day yield change $\Delta \mathbf{y}_t = \mathbf{y}_t - \mathbf{y}_{t-21}$ onto $\mathbf{L}_t$:

$$\mathbf{f}_t = \mathbf{L}_t^T \Delta \mathbf{y}_t \in \mathbb{R}^k \quad (14)$$

and the change is approximated by projecting back:

$$\widetilde{\Delta \mathbf{y}_t} = \mathbf{L}_t \mathbf{f}_t = \mathbf{L}_t \mathbf{L}_t^T \Delta \mathbf{y}_t \quad (15)$$

Since $\mathbf{L}_t$ has orthonormal columns, $\mathbf{L}_t^T \mathbf{L}_t = \mathbf{I}_k$, so $\mathbf{L}_t \mathbf{L}_t^T$ is the orthogonal projection onto the k -dimensional subspace spanned by the retained eigenvectors. The yield curve level is then recovered as:

$$\hat{\mathbf{y}}_{t+h} = \mathbf{y}_t + \mathbf{L}_t \mathbf{f}_{t+h} \quad (16)$$

where $\mathbf{f}_{t+h} = \mathbf{L}_t^T \Delta \mathbf{y}_{t+h}$ are the factors refitted at $t + h$ under the stale loading matrix $\mathbf{L}_t$. Forecasting $h = 21$ days ahead requires that the loading matrix remains approximately stable:

$$\mathbf{L}_{t+h} \approx \mathbf{L}_t \quad (17)$$

This assumption is tested empirically in Section 4.5.

4.4 A Consistent Factor Coordinate System

For the forecasting models developed in subsequent sections, lagged factor values are used as input features. Let $\mathcal{L} = \{0, 22, 43, 64, 85, 106, 127, 148, 169, 190, 211\}$ denote the set of lags in trading days. The specific values are motivated by significant autocorrelation in the factor series, as documented in Section 5.1. For each lag $\ell \in \mathcal{L}$, the 21-day yield change is projected onto the *current* loading matrix $\mathbf{L}_t$:

$$\mathbf{f}_{t-\ell} = \mathbf{L}_t^T \Delta \mathbf{y}_{t-\ell}, \quad \ell \in \mathcal{L} \quad (18)$$

Projecting all lagged observations onto the same $\mathbf{L}_t$ ensures that $\mathbf{f}_{t-\ell}$ across all lags are expressed in a common basis and might not be directly comparable otherwise. The quality of this representation at each lag is measured by:

$$\text{RMSE}_{t-\ell} = \sqrt{\frac{1}{m} \|\Delta \mathbf{y}_{t-\ell} - \mathbf{L}_t \mathbf{L}_t^T \Delta \mathbf{y}_{t-\ell}\|^2}, \quad \ell \in \mathcal{L} \quad (19)$$

This extends the stability assumption (17) beyond the forecast horizon $h = 21$ to the full lag horizon of 211 trading days: reliable factor representations require $\mathbf{L}_t$ to remain a valid basis not only at $t + h$ but also at $t - \ell$ for all $\ell \in \mathcal{L}$. The empirical validity of this assumption across all lag positions and the forecast horizon is tested in Section 4.5. Among the models investigated for the forecasting problem, an LSTM is evaluated in Section ??, where the value of this lag-consistent representation is assessed directly.

4.5 Loading Stability

We test the stability assumptions (17) and (8) empirically across all positions $p \in \mathcal{P}$, where $\mathcal{P} = \mathcal{L} \cup \{+h\}$ covers both the lagged positions and the forecast horizon $h = 21$. For PCA3 and PCA4 the reconstruction RMSE at each position follows (19). The analogous quantity for Nelson-Siegel uses the stale λ_t with factor weights refitted at each position:

$$\text{RMSE}_{t,p}^{\text{NS}} = \sqrt{\frac{1}{m} \|\mathbf{y}_{t+p} - \mathbf{L}(\lambda_t) \boldsymbol{\beta}_{t+p}(\lambda_t)\|^2}, \quad p \in \mathcal{P} \quad (20)$$

Since (20) operates on yield levels while (19) operates on yield changes, the absolute magnitudes across methods are not directly comparable. Both are nonetheless valid stability tests individually: flat profiles across positions indicate that the stale loading structure represents the yield curve consistently over time, including at the forecast horizon $t + h$.

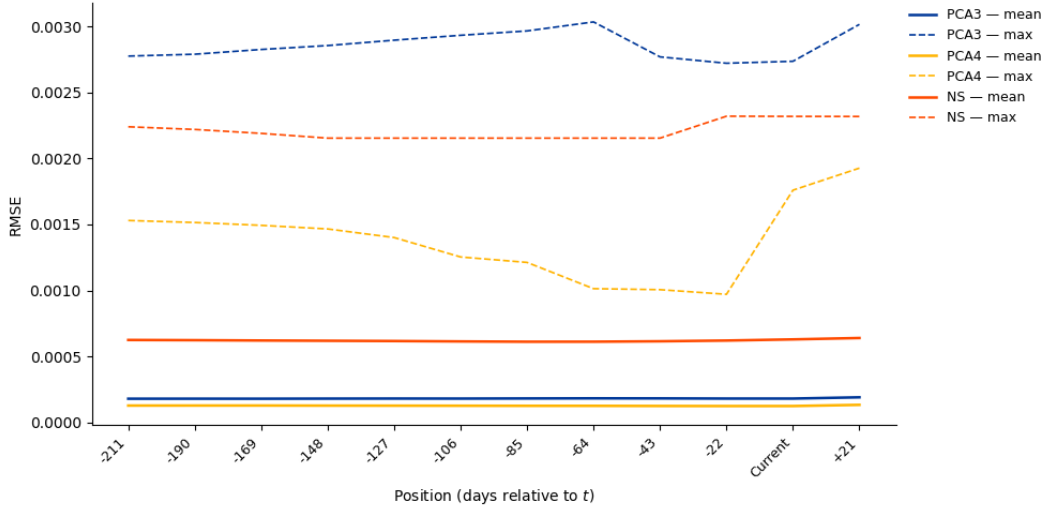


Figure 3: Mean and maximum reconstruction RMSE by position for PCA3, PCA4 and NS, over 6209 observations.

Figure 3 reports the mean and maximum reconstruction RMSE across all 6209 observations,

from position -211 to $+21$ trading days. The mean lines are virtually flat across all positions for all three methods, indicating that on average the stale loading structure represents the yield curve equally well at past positions and at the forecast horizon. The maximum lines, reflecting the worst cases in the distribution, are similarly stable across past positions but show a marginal increase at $+21$ for both PCA specifications, visible also in the distributional results of Appendix A1: the maximum RMSE for PCA4 rises from 0.000971 at position -22 to 0.001927 at $+21$. This marginal deterioration in the tail is expected, as the loading matrix is estimated on data up to t and is applied one month into the future. Crucially, the mean is unaffected, confirming that assumptions (17) and (8) hold in expectation. Among the PCA specifications, PCA4 exhibits a more stable distribution of fitting errors under the loadings assumption than PCA3 across all positions, with consistently lower RMSE indicating that the four-component basis represents yield curve changes more reliably at every position.

4.6 Reconstruction Assessment

Having validated loading stability, we assess how well each method reconstructs the actual yield curve at the forecast horizon $t+h$, as this is the target variable for the forecasting models developed in subsequent sections. Using stale loadings estimated at t , the reconstructed yield curve at $t+h$ follows (16) for PCA and (9) for NS. Reconstruction errors are computed as per-tenor absolute deviations against actual yields at $t+h$, across all 6209 observations.

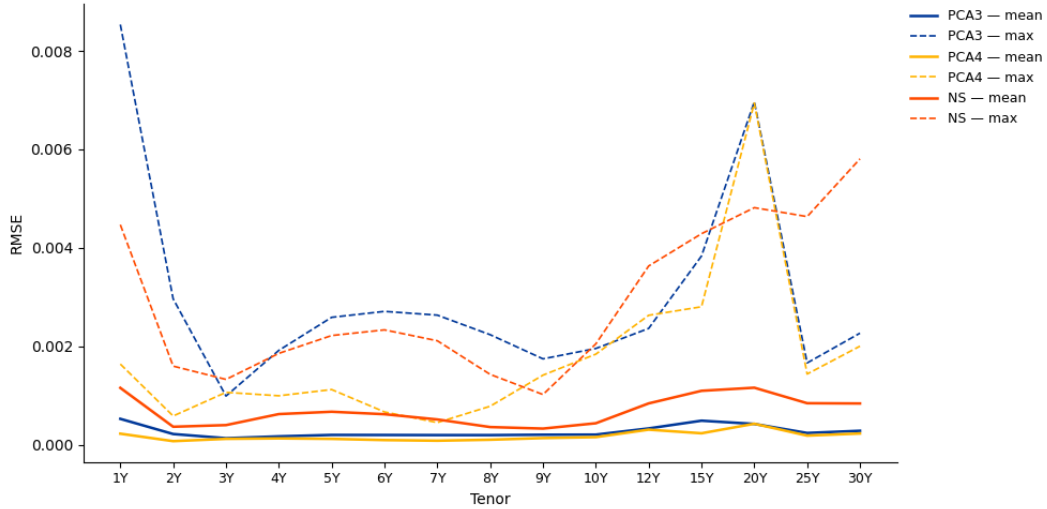
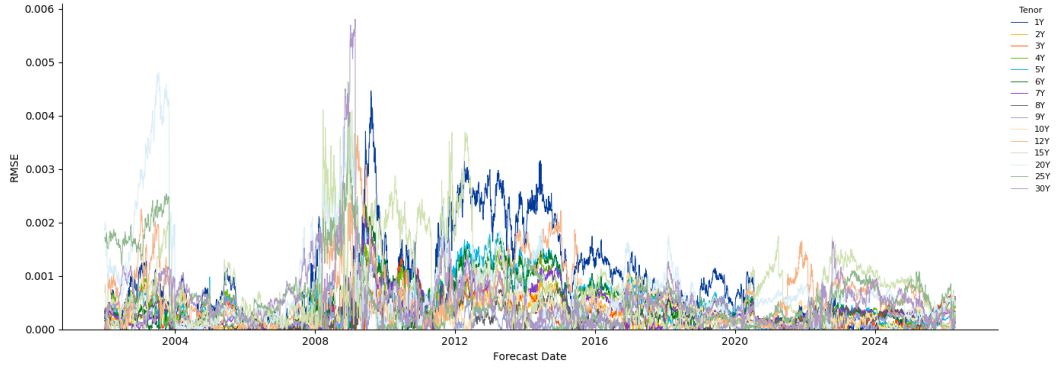
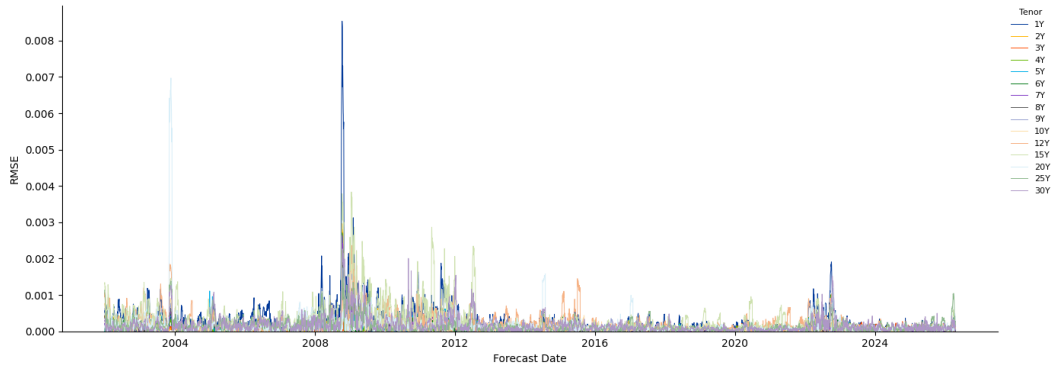


Figure 4: Mean and maximum reconstruction RMSE by tenor for PCA3, PCA4 and NS, over 6209 observations.

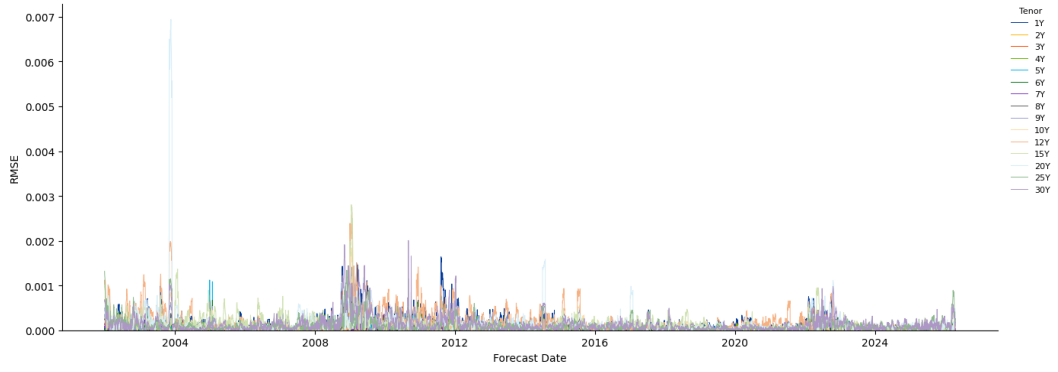
Figure 4 presents mean and maximum reconstruction RMSE across tenors. On the mean, PCA4 achieves the lowest errors across all tenors, followed closely by PCA3, with NS performing considerably worse throughout. Both PCA specifications show mean errors well below NS at every tenor. The maximum lines reflect isolated worst-case observations rather than systematic behaviour: the spike in PCA4 at 20Y corresponds to 2003-11-20, where an unusually large yield movement at 20Y produces a reconstruction error of 0.006941, exceeding the NS maximum of 0.004816 at that tenor at 2003-07-15. Detailed distributional results by tenor are reported in Appendix A2.



(a) NS



(b) PCA3



(c) PCA4

Figure 5: Reconstruction RMSE per tenor over time for NS, PCA3 and PCA4.

Figure 5 plots reconstruction RMSE per tenor over time. NS exhibits persistently elevated and broadly distributed errors across the full sample, with stress periods such as the 2008 financial crisis and its aftermath producing the largest spikes. Both PCA specifications show considerably lower baseline errors, with the largest deviations concentrated in stress episodes. For PCA3, the largest outliers are concentrated at the short end of the curve during the 2008 financial crisis, reflecting that three components are insufficient to capture short-end dynamics under irregular yield curve shapes. The sovereign debt crisis extending into 2012 and the Covid crisis around 2020 are also visible as elevated periods in all three methods, although for PCA 3 and 4 at considerably lower magnitudes. PCA4 reduces short-end errors in comparison to PCA3. Across tenors and time periods, PCA4 demonstrates the most reliable reconstruction.

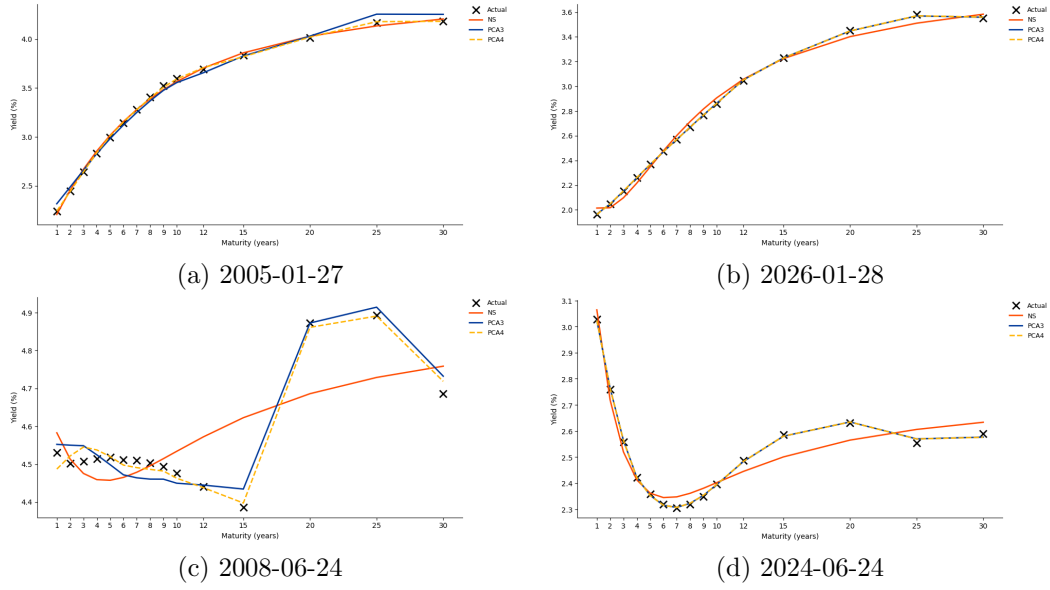


Figure 6: Yield curve fits for selected dates.

Figure 6 illustrates the fit qualitatively across four yield curve environments. In a normal upward sloping environment (panel a), NS fits reasonably well but PCA4 tracks the curve more precisely. In the most recent observation (panel b), PCA3 and PCA4 are nearly indistinguishable and both outperform NS. Under the irregular 2008 hump shape (panel c) and the 2024 inverted curve (panel d), NS fails to capture the shape entirely, while PCA4 continues to track the actual curve closely. These findings motivate our choice to base our forecasting framework on PCA4.

5 Forecasting Principal Components with VAR

5.1 Autocorrelation Analysis and Lag Selection

For each component $k \in \{1, 2, 3, 4\}$, the target series is the 21-trading-day change in the k -th principal component:

$$\Delta f_{k,t} = f_{k,t+h} - f_{k,t}, \quad h = 21 \quad (21)$$

To identify the dependence structure, we compute the autocorrelation function (ACF) and partial autocorrelation function (PACF). The ACF at lag ℓ is:

$$\hat{\rho}(\ell) = \frac{\sum_{t=\ell+1}^N (\Delta f_{k,t} - \overline{\Delta f_k})(\Delta f_{k,t-\ell} - \overline{\Delta f_k})}{\sum_{t=1}^N (\Delta f_{k,t} - \overline{\Delta f_k})^2} \quad (22)$$

The PACF at lag ℓ isolates the direct effect of lag ℓ after conditioning on all intermediate lags $1, \dots, \ell - 1$, obtained via the Yule-Walker equations Lütkepohl (2005). A sharp PACF cutoff with gradual ACF decay is the identifying signature of an autoregressive process. Significance is assessed against the 95% confidence bound $\pm 1.96/\sqrt{N}$.

Because $\Delta f_{k,t}$ is a cumulative h -period change, consecutive observations share $h - 1 = 20$ overlapping days, mechanically contaminating the ACF and PACF for lags $\ell < h = 21$. The first interpretable lag is therefore $\ell^* = h + 1 = 22$, coinciding with the first element of $\mathcal{L}$ introduced in Section 4.4.

Figure 7 presents the results. The PACF shows a large positive spike at $\ell^* = 22$ for all four components, with values between 0.46 and 0.61, confirming strong direct mean-reversion at the first valid lag. Significant partial autocorrelations persist to lag 211 for f_1 , to approximately lag 127 for f_2 and f_3 , and with intermediate persistence for f_4 . The gradual ACF decay without a clean cutoff is consistent with an autoregressive rather than moving-average structure. Based on these findings, the forecasting models use $\mathcal{L}_{\text{VAR}} = \{22, 43, 64, 85, 106, 127, 148, 169, 190, 211\}$, capturing the full persistence horizon of f_1 without unnecessarily extending the model. Full ACF and PACF values at all lags in $\mathcal{L}_{\text{VAR}}$ are reported in Appendix A3.

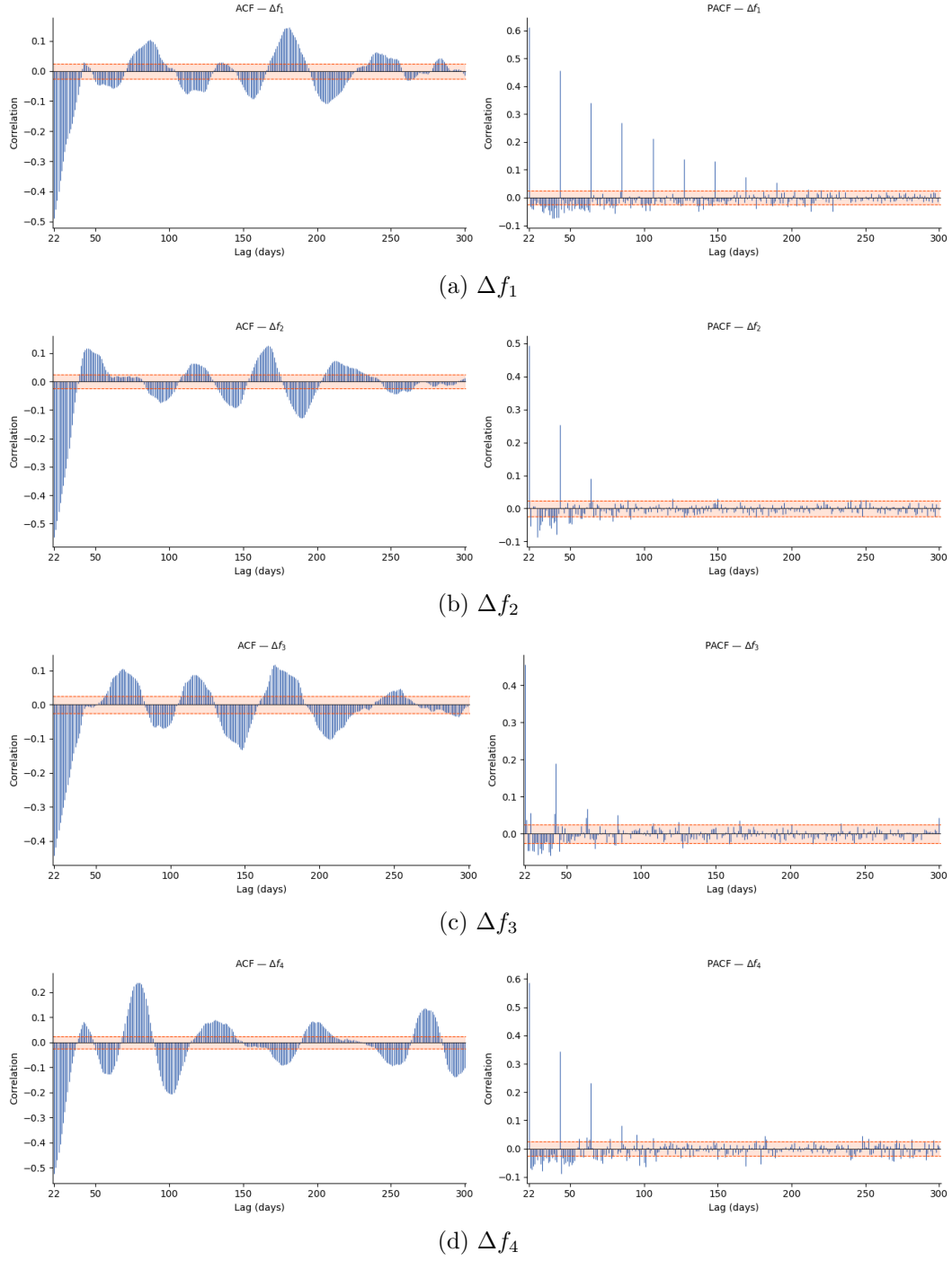


Figure 7: ACF (left) and PACF (right) of 21-day principal component changes $\Delta f_{k,t}$ for $k \in \{1, 2, 3, 4\}$. Dashed lines indicate 95% confidence bounds $\pm 1.96/\sqrt{N}$.

5.2 VAR Specification and In-Sample Results

The autocorrelation analysis of Section 5.1 motivates a vector autoregression (VAR) over the four principal component changes $\Delta \mathbf{f}_t = (\Delta f_{1,t}, \Delta f_{2,t}, \Delta f_{3,t}, \Delta f_{4,t})^\top$. Following Diebold and Li (2006), who establish the VAR as the natural benchmark for factor-based yield curve forecasting, we model each component jointly with the others using the lag set $\mathcal{L}$. The system is:

$$\Delta f_{k,t} = c_k + \sum_{\ell \in \mathcal{L}_{\text{VAR}}} \sum_{j=1}^4 \phi_{k,j}^{(\ell)} \Delta f_{j,t-\ell} + \varepsilon_{k,t}, \quad k \in \{1, 2, 3, 4\} \quad (23)$$

where c_k is an intercept, $\phi_{k,j}^{(\ell)}$ is the coefficient on the ℓ -th lag of component j in equation k , and $\varepsilon_{k,t}$ is the equation error. Each equation contains $|\mathcal{L}_{\text{VAR}}| \times 4 = 40$ regressors and is estimated separately by OLS:

$$\hat{\boldsymbol{\theta}}_k = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}_k^* \quad (24)$$

where $\mathbf{X}$ is the $N \times 41$ design matrix of lagged regressors plus intercept and $\mathbf{y}_k^* \in \mathbb{R}^N$ is the vector of observations of $\Delta f_{k,t}$.

Table 3: VAR in-sample fit statistics.

PC	N	R^2	RMSE	AIC	BIC	DoF
PC1	5998	0.4683	0.007534	-41537.15	-41262.49	5957
PC2	5998	0.4668	0.002498	-54779.91	-54505.24	5957
PC3	5998	0.3362	0.001242	-63163.09	-62888.42	5957
PC4	5998	0.4581	0.000867	-67470.89	-67196.22	5957

Table 3 reports the in-sample fit statistics. All four equations achieve substantial explanatory power, with R^2 ranging from 0.34 for f_3 to 0.47 for f_1 and f_2 , confirming that the lag structure identified from the ACF and PACF carries genuine predictive content. Full per-equation coefficient tables with standard errors, t -statistics, and significance levels are provided in Appendix A4.

Table 4: Cross-PC significance matrix. Each entry reports the number of statistically significant lags out of 10 at $\alpha = 0.05$.

Equation \ Predictor	PC1	PC2	PC3	PC4
PC1	10/10	7/10	9/10	7/10
PC2	8/10	8/10	8/10	8/10
PC3	8/10	4/10	7/10	8/10
PC4	9/10	8/10	6/10	8/10

Table 4 summarises the number of statistically significant cross-lags per predictor-equation pair. The principal components, while orthogonal by construction in the cross-sectional sense, are not dynamically independent. Lagged values of f_1 , the parallel shift factor carrying the largest share of yield curve variance, are significant in at least 8 out of 10 lags in every non-own equation, making it the strongest cross-predictor in the system. This is economically interpretable: parallel level shifts propagate into slope and curvature dynamics with a delay. The implication is important for model specification: a univariate autoregressive model for each component would discard this cross-predictive information, and the VAR framework is therefore the appropriate linear benchmark.

5.3 Rolling VAR and Out-of-Sample Results

To eliminate look-ahead bias, we implement a rolling walk-forward procedure. At each prediction date t , the VAR is estimated on the τ most recent observations ending at $t - h$ and used to predict $\Delta f_{k,t}$:

$$\widehat{\Delta f}_{k,t} = \hat{c}_k^{(t)} + \sum_{\ell \in \mathcal{L}_{\text{VAR}}} \sum_{j=1}^4 \hat{\phi}_{k,j}^{(\ell,t)} \Delta f_{j,t-\ell} \quad (25)$$

where the superscript (t) denotes that coefficients are re-estimated at each step using only data in $[t - h - \tau, t - h]$. The directional prediction is:

$$\hat{d}_{k,t} = \mathbb{1} \left[\widehat{\Delta f}_{k,t} > 0 \right] \quad (26)$$

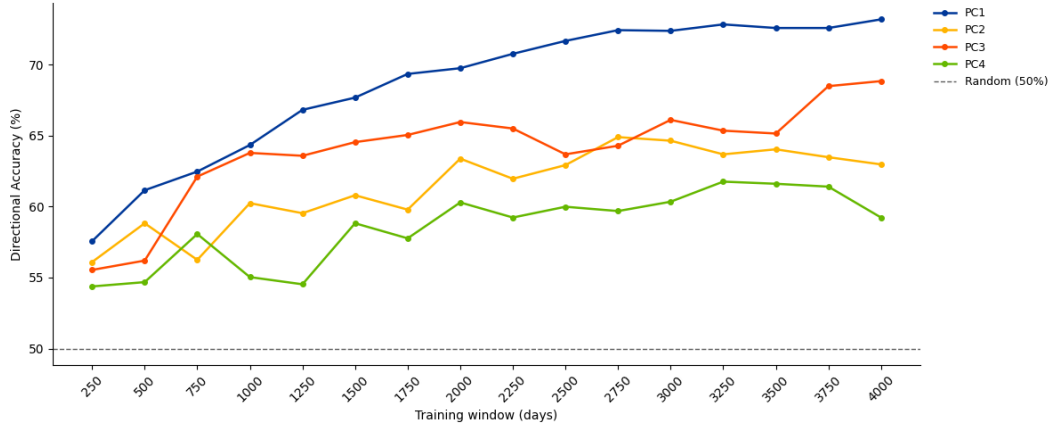
evaluated against the realised direction $d_{k,t} = \mathbb{1}[\Delta f_{k,t} > 0]$. Directional accuracy is then defined as:

$$\text{DA}_k = \frac{1}{N} \sum_{t=1}^N \mathbb{1} \left[\hat{d}_{k,t} = d_{k,t} \right] \quad (27)$$

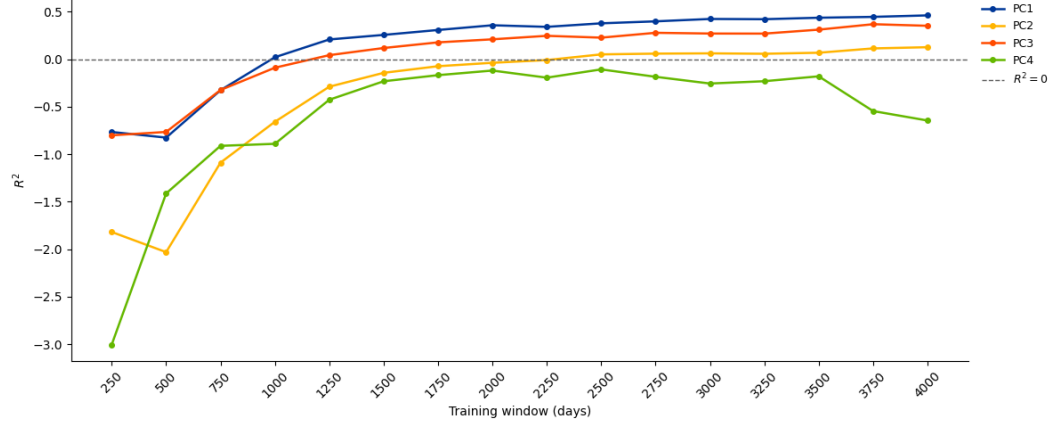
5.3.1 Training Window Selection

The choice of τ involves a trade-off between parameter stability and estimation precision: longer windows reduce parameter uncertainty across 40 regressors per equation but may incorporate observations that are no longer representative. We evaluate candidate windows $\tau \in \{250, 500, \dots, 4000\}$ on a common evaluation sample of $n = 1,977$ observations from 2018-06-22 to 2026-03-13.

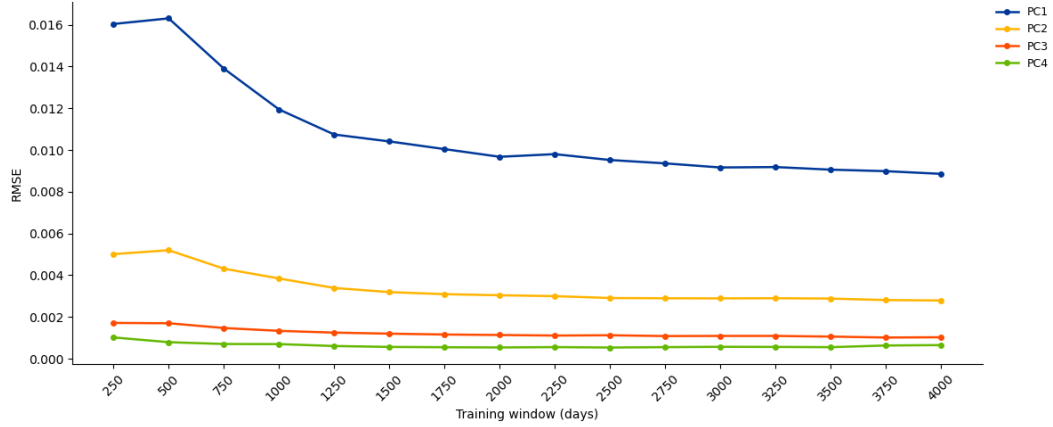
Figure 8 presents the results. Three patterns emerge. First, directional accuracy increases monotonically with τ for f_1 , stabilising around $\tau = 2,500$, while improvements for f_2 , f_3 , and f_4 are less systematic beyond $\tau = 2,000$. Second, OOS R^2 is deeply negative at small windows and converges to positive values from $\tau \approx 1,250$ for f_1 and f_3 , while remaining negative for f_4 throughout. Third, RMSE decreases steadily with τ for all components up to approximately $\tau = 3,000$, with only marginal improvements at larger windows. We select $\tau = 3,000$ as a defensible choice for our forecasting framework: accuracy and RMSE are near their best values across components, while retaining $N = 2,977$ out-of-sample predictions. Extending to $\tau = 4,000$ yields negligible further improvement in accuracy or RMSE while substantially reducing the evaluation horizon.



(a) Directional Accuracy



(b) R^2



(c) RMSE

Figure 8: Rolling VAR directional accuracy, OOS R^2 and RMSE as a function of training window size τ , evaluated on the common sample ($n = 1,977$).

5.3.2 Out-of-Sample Results

Table 5: OOS Rolling VAR, $\tau = 3,000$.

PC	N	Accuracy	R^2	RMSE	AIC	BIC
PC1	2977	72.09%	0.4092	0.008325	-19980.27	-19734.33
PC2	2977	65.87%	0.0532	0.002730	-26619.07	-26373.13
PC3	2977	64.43%	0.2628	0.001032	-32411.67	-32165.72
PC4	2977	62.08%	-0.2065	0.000567	-35974.91	-35728.97

Table 6: OOS Rolling VAR forecast residual statistics, $\tau = 3,000$.

PC	Mean	Std. Dev.	Min	Q25	Q75	Max
PC1	0.000158	0.008324	-0.036123	-0.004629	0.004874	0.036659
PC2	0.000046	0.002729	-0.012505	-0.001365	0.001233	0.015597
PC3	0.000037	0.001031	-0.004325	-0.000547	0.000644	0.005226
PC4	0.000036	0.000566	-0.002095	-0.000271	0.000317	0.003605

A notable finding is that f_3 achieves both higher directional accuracy (64.43%) and substantially higher OOS R^2 (0.263) than f_2 (65.87%, 0.053), despite f_2 explaining more yield curve variance in-sample: curvature changes appear more linearly predictable in magnitude than slope changes. For f_4 the VAR achieves 62.08% directional accuracy, though the negative OOS R^2 of -0.21 indicates that the magnitude of f_4 changes is difficult to forecast linearly, even where direction is captured. Residual means across all components are negligibly close to zero, confirming the absence of systematic bias. The VAR thus establishes the linear benchmark for yield curve factor forecasting. The following section investigates whether an LSTM can capture non-linear dynamics in the factor series and improve upon this baseline.

6 Forecasting Principal Components with LSTM

6.1 Feature Construction

Two categories of input features are distinguished: sequential features, which vary across timesteps and are processed by the recurrent layers, and static features, which are constant across timesteps and concatenated after the recurrent pass.

6.1.1 Sequential Features

For each $\ell \in \mathcal{L}$, a ten-dimensional feature vector is constructed. Four features are the factor coordinates $\mathbf{f}_{t-\ell}$ as defined in (18), and one is the reconstruction error $\text{RMSE}_{t-\ell}$ as defined in (19); both were introduced in Section 4.4.

The implied volatility of the $3\text{M} \times 10\text{Y}$ ATM EUR swaption, expressed in basis points, is transformed into a 21-day percentage return series, $\text{IV}_{t-\ell} = (\text{IV}_{t-\ell} - \text{IV}_{t-\ell-21}) / \text{IV}_{t-\ell-21}$, and used as a sixth sequential feature in LSTM-VAR-FULL-EIG-IV only. This provides a signal about market-implied rate uncertainty at each lag position.

The remaining four features are the lagged factor differences $\Delta f_{k,t-\ell}$ for $k \in \{1, 2, 3, 4\}$, as defined in (21). At $\ell = 0$ these are set to zero:

$$\Delta f_{k,t} = 0, \quad \ell = 0 \quad (28)$$

since $\Delta f_{k,t} = f_{k,t+h} - f_{k,t}$ involves the forecast target and is unavailable at prediction time. The sequential input is thus a tensor of shape $11 \times p$, where $p \in \{4, 5, 10\}$ depending on the variant; see Table 7.

6.1.2 Static Features

The four daily eigenvalues $\boldsymbol{\nu}_t^\delta = (\nu_{1,t}^\delta, \dots, \nu_{4,t}^\delta)$ from $\hat{\boldsymbol{\Sigma}}_t^\delta$, as introduced in (10), are fed as static context. These capture the time-varying explained variance per component and provide a compact summary of the prevailing yield curve volatility regime.

6.1.3 Feature Standardisation

All features are standardised by a `StandardScaler` fitted on the training window $[t-h-\tau, t-h)$ and applied to the prediction observation. Concretely, for a feature with in-window mean μ and standard deviation σ :

$$\tilde{x} = \frac{x - \mu}{\sigma} \quad (29)$$

Fitting exclusively on the training window prevents data leakage: using any information from t or beyond would implicitly reveal the distributional location of the quantity being forecast. Rolling re-standardisation also stabilises the input scale across volatility regimes.

6.2 Model Variants

Four variants are evaluated. Table 7 summarises their feature sets and architectures; Figure 9 shows the full graph of the largest variant.

Table 7: LSTM variant feature sets and architectures.

Variant	Sequential ($11 \times p$)	Static	Head
LSTM-VAR	$\Delta f_{k,t-\ell}$ $p = 4$	—	Dense(4)
LSTM-FULL	$\mathbf{f}_{t-\ell}$, $\text{RMSE}_{t-\ell}$ $p = 5$	—	Dense(4)
LSTM-FULL-EIG	$\mathbf{f}_{t-\ell}$, $\text{RMSE}_{t-\ell}$ $p = 5$	$\boldsymbol{\nu}_t^\delta$	Dense(16) + Dense(4)
LSTM-VAR-FULL-EIG-IV	all 10 features $p = 10$	$\boldsymbol{\nu}_t^\delta$	Dense(16) + Dense(4)

All variants share the recurrent backbone: two stacked LSTM layers of 50 and 20 units. LSTM-VAR and LSTM-FULL pass the final hidden state directly to the output layer. The two variants with static input concatenate $\boldsymbol{\nu}_t^\delta$ with the hidden state, pass it through a Dense(16, ReLU) layer with Dropout($p = 0.2$), and then to the output.

LSTM-VAR uses only $\Delta f_{k,t-\ell}$, the same information as the VAR in (23), making it the direct nonlinear benchmark. LSTM-FULL replaces these with the coordinate-consistent factor representation from Section 4.4, adding $\text{RMSE}_{t-\ell}$ as a quality signal. LSTM-FULL-EIG augments this with the eigenvalues as static regime context. LSTM-VAR-FULL-EIG-IV combines all information sources, adding both $\Delta f_{k,t-\ell}$ and $\text{IV}_{t-\ell}$ to the sequential input.

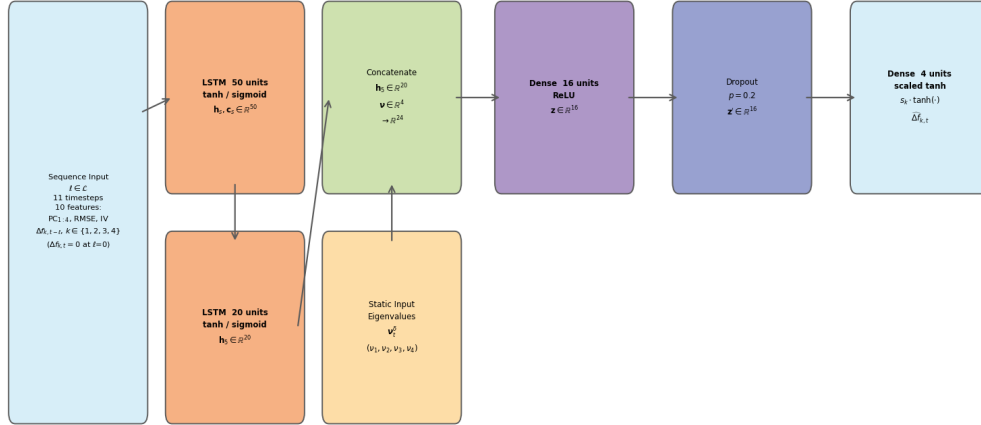


Figure 9: Architecture of LSTM-VAR-FULL-EIG-IV. Variants without static input omit the concatenation and Dense(16) head; see Table 7.

6.3 Architecture and Training

6.3.1 LSTM Layers

At each timestep $s \in \{1, \dots, S\}$, with $S = 11$, the LSTM updates its hidden state $\mathbf{h}_s \in \mathbb{R}^d$ and cell state $\mathbf{c}_s \in \mathbb{R}^d$ via:

$$\mathbf{f}_s = \sigma(\mathbf{W}_f[\mathbf{h}_{s-1}, \mathbf{x}_s] + \mathbf{b}_f) \quad (30)$$

$$\mathbf{i}_s = \sigma(\mathbf{W}_i[\mathbf{h}_{s-1}, \mathbf{x}_s] + \mathbf{b}_i) \quad (31)$$

$$\tilde{\mathbf{c}}_s = \tanh(\mathbf{W}_c[\mathbf{h}_{s-1}, \mathbf{x}_s] + \mathbf{b}_c) \quad (32)$$

$$\mathbf{c}_s = \mathbf{f}_s \odot \mathbf{c}_{s-1} + \mathbf{i}_s \odot \tilde{\mathbf{c}}_s \quad (33)$$

$$\mathbf{o}_s = \sigma(\mathbf{W}_o[\mathbf{h}_{s-1}, \mathbf{x}_s] + \mathbf{b}_o) \quad (34)$$

$$\mathbf{h}_s = \mathbf{o}_s \odot \tanh(\mathbf{c}_s) \quad (35)$$

where σ denotes the sigmoid function and $\odot$ elementwise multiplication. The forget gate $\mathbf{f}_s$ controls how much of the previous cell state is retained, the input gate $\mathbf{i}_s$ determines what new information enters, and the output gate $\mathbf{o}_s$ filters what is passed forward as the hidden state. The cell state $\mathbf{c}_s$ therefore acts as a selective memory that can propagate information across the full lag horizon without the vanishing gradient problem of plain RNNs.

Two LSTM layers are stacked: the first with $d = 50$ units processes the raw temporal patterns across the 11 timesteps, the second with $d = 20$ units compresses this into a lower-dimensional representation before the feedforward head. Both layers use `tanh` cell activation and `sigmoid` recurrent activation, consistent with (30) to (35). All states are initialised at zero, $\mathbf{h}_0 = \mathbf{c}_0 = \mathbf{0}$, since no prior information about the sequence is available at the start of each window.

6.3.2 Feedforward Head

For variants without static input (LSTM-VAR, LSTM-FULL), the final hidden state $\mathbf{h}_S \in \mathbb{R}^{20}$ is passed directly to the output layer. For variants with static input (LSTM-FULL-EIG, LSTM-VAR-FULL-EIG-IV), $\mathbf{h}_S$ is concatenated with the standardised eigenvalues $\tilde{\boldsymbol{\nu}}_t^\delta \in \mathbb{R}^4$ to form a vector in $\mathbb{R}^{24}$, which is passed through:

$$\mathbf{z} = \text{ReLU}(\mathbf{W}_1[\mathbf{h}_S, \tilde{\boldsymbol{\nu}}_t^\delta] + \mathbf{b}_1), \quad \text{ReLU}(x) = \max(0, x) \quad (36)$$

followed by Dropout with $p = 0.2$, which randomly zeroes activations during training to reduce overfitting.

6.3.3 Output Layer and Scaled Tanh

All four components are predicted simultaneously. The raw output of the final Dense(4) layer is passed through a scaled tanh:

$$\widehat{\Delta f}_{k,t+h} = s_k \cdot \tanh(\mathbf{w}_k^\top \mathbf{z}'), \quad s_k = \max\left(\max_t \Delta f_{k,t}, -\min_t \Delta f_{k,t}\right) \quad (37)$$

where $\mathbf{z}'$ denotes the penultimate activation and s_k is the maximum absolute value of $\Delta f_{k,t}$ over the training window $[t-h-\tau, t-h]$. The targets are normalised to $[-1, 1]$ before training via $\Delta f_{k,t}/s_k$. This bounds predictions within the observed range of factor movements and is recomputed at each rolling step.

6.3.4 Training Procedure

All variants are estimated using a rolling walk-forward procedure identical to that of the VAR in Section 5.3. At each prediction date t , a fresh model is initialised and trained from scratch on the $\tau = 3000$ most recent observations in $[t - h - \tau, t - h)$, then used to produce a single out-of-sample prediction for $\widehat{\Delta f}_{k,t+h}$. No parameters are carried forward across windows.

Each rolling step requires a full training run, and with $N = 2977$ prediction dates the total computation amounts to thousands of independent training jobs. The models were implemented in TensorFlow and executed on a 16-core, 64 GB Cloudera AI environment with 16 parallel workers, each responsible for an independent subset of rolling windows. This design is nonetheless the natural evaluation framework: it mirrors exactly how a practitioner would deploy the model, with access to the most recent 3000 observations and a requirement to produce one forecast for the next trading month.

The loss function is mean absolute error between the scaled predictions and the normalised targets $\Delta f_{k,t}/s_k$. Training uses the Adam optimiser with learning rate $\alpha = 0.001$, batch size 32, and a maximum of 500 epochs. A `ReduceLROnPlateau` callback halves the learning rate after 10 epochs without improvement in training loss, with a floor of 10^{-5} . Early stopping with patience 40 and `restore_best_weights=True` terminates training when training loss ceases to improve, returning the lowest-loss checkpoint. Since the model is evaluated entirely out-of-sample, the training window $[t - h - \tau, t - h)$ never overlapping with the prediction date t , the rolling design itself provides the holdout and the models will be validated through their out of sample predictions.

6.4 Out-of-Sample Results

Table 8 reports directional accuracy, OOS R^2 , and RMSE for all variants alongside the VAR benchmark from Table 5. Full per-variant performance and residual tables are provided in Appendix A5.

Table 8: OOS comparison: VAR and LSTM variants, $\tau = 3,000$, $N = 2977$.

Variant	Accuracy (%)				R^2				RMSE ($\times 10^3$)			
	PC1	PC2	PC3	PC4	PC1	PC2	PC3	PC4	PC1	PC2	PC3	PC4
VAR	72.09	65.87	64.43	62.08	0.409	0.053	0.263	-0.207	8.33	2.73	1.03	0.57
LSTM-VAR	64.70	56.73	59.05	57.10	0.095	-0.248	-0.155	-0.657	10.30	3.13	1.29	0.67
LSTM-FULL	70.27	68.26	65.97	65.50	0.236	0.158	0.096	-0.129	9.47	2.58	1.14	0.55
LSTM-FULL-EIG	69.70	65.57	65.77	55.59	0.257	0.137	0.176	-0.112	9.34	2.61	1.09	0.54
LSTM-VAR-FULL-EIG-IV	71.08	67.48	65.40	60.36	0.318	0.225	0.198	0.003	8.95	2.47	1.08	0.52

6.4.1 LSTM-VAR versus VAR

LSTM-VAR operates on identical information to the VAR and performs worse on all three metrics across every component. PC1 accuracy falls from 72.09% to 64.70%, R^2 from 0.41 to 0.09, and RMSE ($\times 10^3$) widens from 8.33 to 10.30. The negative R^2 values for PC2 through PC4 indicate that the model's continuous predictions are worse than a simple mean forecast in terms of magnitude, even where directional accuracy remains above 50%. The mean-reversion structure identified in Section 5.1 is well captured linearly by the VAR, and

the specific LSTM architecture chosen here does not replicate that structure when restricted to the same inputs.

6.4.2 LSTM-FULL versus LSTM-VAR and VAR

Replacing the raw factor differences with the coordinate-consistent representation substantially improves all metrics. Directional accuracy exceeds the VAR for PC2 (68.26% vs 65.87%), PC3 (65.97% vs 64.43%), and PC4 (65.50% vs 62.08%), and RMSE improves for PC2 and PC4 relative to both the VAR and LSTM-VAR. The R^2 for PC2 rises from 0.053 to 0.158, a meaningful improvement in magnitude prediction, while PC4 remains negative but improves from -0.207 under the VAR to -0.129 . PC1 remains below the VAR on both accuracy and R^2 , consistent with the level factor being the most linearly predictable component.

The gains for PC2 and PC4 in particular suggest that nonlinear dynamics may be present for these factors that the VAR cannot capture. For PC4, the R^2 remaining negative indicates that magnitude prediction is still difficult. Taken together, LSTM-FULL provides evidence that the coordinate-consistent representation from Section 4.4 gives the LSTM access to genuinely incremental information.

6.4.3 LSTM-FULL-EIG versus LSTM-FULL

Adding eigenvalues as static context produces mixed results. R^2 improves for PC1 (0.236 to 0.257) and PC3 (0.096 to 0.176), and RMSE is marginally lower across most components. However, directional accuracy declines for PC2 (68.26% to 65.57%) and PC4 (65.50% to 55.59%), with the latter dropping below the VAR benchmark. Where accuracy falls while R^2 and RMSE improve only marginally, the model sacrifices directional correctness without a compensating gain in magnitude prediction. The eigenvalues do not add consistent value across components.

6.4.4 Combined Variant

LSTM-VAR-FULL-EIG-IV achieves the strongest results across the magnitude metrics. It records the best R^2 for PC2 (0.225) and PC4 (0.003), and the lowest RMSE for PC2 (2.47) and PC4 (0.52). Notably, PC4 is the only component where any LSTM variant achieves a positive R^2 , indicating that combining all information sources finally enables the model to predict the magnitude of higher-order factor movements above the mean forecast baseline. On PC1, it closely approaches the VAR in both accuracy (71.08% vs 72.09%) and RMSE (8.95 vs 8.33), though the VAR retains the best R^2 for the level factor.

Directional accuracy is more mixed: LSTM-FULL retains the highest accuracy for PC2 and PC4, suggesting that adding the raw factor differences and implied volatility in the combined variant improves magnitude prediction at some cost to directional sharpness. The inclusion of swaption implied volatility as a sequential feature is unique to this variant and likely contributes to the PC4 R^2 improvement, though its individual effect is not isolated here. Across all metrics and components together, LSTM-VAR-FULL-EIG-IV represents the most balanced specification and will be carried forward to the yield curve reconstruction in Section 7.

7 Yield Curve Reconstruction

7.1 From Factor Forecasts to Yield Levels

The factor predictions $\widehat{\Delta f}_{k,t+h}$ from Section 6.4 are mapped back to yield curve levels via (16):

$$\widehat{\mathbf{y}}_{t+h} = \mathbf{y}_t + \mathbf{L}_t \widehat{\Delta \mathbf{f}}_{t+h}$$

Four specifications are evaluated over the common out-of-sample window of $N = 2977$ observations from 2014-08-22 to 2026-04-20: the rolling VAR, the LSTM-VAR-FULL-EIG-IV model, a hybrid model combining VAR predictions for PC1 and PC3 with LSTM predictions for PC2 and PC4, and the random walk benchmark $\widehat{\mathbf{y}}_{t+h} = \mathbf{y}_t$. The hybrid specification is motivated by the factor-level results in Section 6.4.

7.2 Results

Table 9 reports RMSE and directional accuracy by tenor for all four specifications. Full residual statistics are provided in Appendix A6.

Table 9: Yield curve reconstruction performance by tenor, $N = 2977$.

Tenor	RMSE ($\times 10^3$)				Dir. Acc. (%)			
	RW	VAR	LSTM	Hybrid	RW	VAR	LSTM	Hybrid
1Y	1.705	2.201	2.508	2.284	0.03	54.32	50.62	53.54
2Y	1.774	2.275	2.607	2.391	0.03	56.20	51.09	53.68
3Y	1.845	2.375	2.716	2.495	0.10	57.14	53.04	54.52
4Y	1.895	2.466	2.809	2.578	0.03	56.53	53.28	53.98
5Y	1.927	2.539	2.879	2.634	0.00	55.53	53.95	54.25
6Y	1.950	2.593	2.922	2.665	0.10	54.79	53.38	53.85
7Y	1.969	2.641	2.952	2.686	0.10	53.85	53.41	53.71
8Y	1.992	2.694	2.986	2.712	0.03	53.75	53.44	53.85
9Y	2.016	2.747	3.021	2.742	0.07	54.01	53.98	53.78
10Y	2.031	2.791	3.053	2.769	0.03	53.88	53.75	54.08
12Y	2.091	2.904	3.122	2.846	0.00	52.91	53.51	53.51
15Y	2.135	2.958	3.188	2.904	0.03	53.48	52.47	53.41
20Y	2.197	3.080	3.282	2.999	0.00	51.86	52.00	52.23
25Y	2.177	3.066	3.260	2.973	0.00	51.53	51.66	52.23
30Y	2.179	3.050	3.246	2.957	0.03	52.00	51.97	52.60

7.2.1 Random Walk Benchmark

The random walk achieves the lowest RMSE at every tenor, ranging from 1.705 at 1Y to 2.197 at 20Y. All three forecasting models produce higher absolute errors, with negative R^2 throughout. This is consistent with the findings of Diebold and Li (2006), who document that the random walk is an exceptionally difficult benchmark to beat at the one-month horizon and that outperformance of factor models concentrates at longer horizons of six to

twelve months. At $h = 21$ trading days, yield changes are small relative to yield levels and contain substantial noise, so the no-change prediction minimises squared error mechanically. The random walk’s near-zero directional accuracy ($\approx 0\%$) reflects the fact that it predicts no movement and therefore has no directional content whatsoever.

7.2.2 VAR versus LSTM

Among the forecasting models, the VAR dominates at the short end on both RMSE and directional accuracy, consistent with PC1 being the most linearly predictable factor. The pure LSTM underperforms the VAR across all tenors on RMSE and at the short end on directional accuracy, and does not represent a standalone improvement over the linear benchmark at the yield curve level.

7.2.3 Hybrid Model

The hybrid model recovers the VAR’s short-end accuracy while improving upon it at longer tenors, where substituting the LSTM’s PC2 and PC4 predictions proves beneficial. From roughly 8Y onwards the hybrid achieves better directional accuracy and lower RMSE than both the VAR and the pure LSTM, as Table 9 shows.

8 Conclusion

This thesis develops a framework for forecasting the German zero-coupon yield curve by combining dimensionality reduction, linear factor forecasting, and nonlinear sequence modelling. The analysis proceeds in three stages: fitting the yield curve, forecasting its principal components, and reconstructing yield levels from those forecasts.

Among the two fitting approaches evaluated, PCA4 consistently outperforms both PCA3 and Nelson-Siegel across all maturities and yield curve regimes. The fourth component materially reduces short-end reconstruction errors and provides a more stable basis across the full lag horizon required by the forecasting models. The Nelson-Siegel model, despite its economic interpretability, fails to capture irregular yield curve shapes. These findings motivate the choice of PCA4 as the foundation for the forecasting framework.

The VAR analysis reveals two results that are not immediately obvious. First, although the four principal components are orthogonal by construction, their lagged values are dynamically interdependent: the level factor PC1 is a significant predictor in the equations for all other components, confirming that parallel yield curve shifts propagate into slope and curvature dynamics with a delay. A univariate autoregression per component would discard this information, making the VAR the appropriate linear specification. Second, the out-of-sample results show that the VAR achieves meaningful directional accuracy across all four components.

The LSTM investigation shows that the choice of input representation matters substantially. Feeding the LSTM the same raw lagged factor differences as the VAR produces inferior results on all metrics, suggesting that the linear mean-reversion structure is not readily extracted by the chosen architecture without the appropriate feature encoding. Replacing those inputs with the coordinate-consistent factor representation developed in Section 4.4 substantially improves performance, particularly for PC2 and PC4, where the LSTM surpasses the VAR on directional accuracy and R^2 . This suggests that nonlinear dynamics are

present in the slope and higher-order components that the VAR cannot fully capture, while the level factor PC1 remains best described by the linear model. The combined variant LSTM-VAR-FULL-EIG-IV, which additionally incorporates 21-day percentage returns of $3M \times 10Y$ ATM normal swaption implied volatility as a sequential feature, achieves the only positive OOS R^2 observed for PC4 across any specification. That the best-performing variant includes implied volatility suggests that market-implied rate uncertainty carries incremental information about higher-order factor dynamics beyond what is contained in the factor history alone, though isolating its individual contribution relative to the other added features remains a direction for further investigation.

At the yield curve reconstruction level, no forecasting model beats the random walk over the 21-trading-day horizon on RMSE. This is consistent with Diebold and Li (2006), who document that factor model outperformance concentrates at horizons of six months and beyond, where mean-reversion in factors becomes exploitable over levels noise. The hybrid model, which combines VAR predictions for PC1 and PC3 with LSTM predictions for PC2 and PC4, does not improve on the random walk in terms of RMSE but achieves the strongest directional accuracy at longer tenors. From roughly 8Y to 30Y, the hybrid outperforms both the pure VAR and pure LSTM on direction and RMSE, demonstrating that the complementary strengths of the two models can be combined productively at the yield curve level.

The central finding of this thesis is therefore that nonlinear dynamics in the principal components of the German yield curve seem present and partially capturable, but translating factor-level forecasting gains into yield curve level improvements over a one-month horizon remains difficult. The coordinate-consistent representation introduced here provides a principled way to encode temporal structure for sequence models, and the hybrid architecture offers a practical approach to combining linear and nonlinear forecasts. Extending the evaluation to longer horizons, where mean-reversion in factors is more pronounced, and investigating whether the LSTM’s directional advantage for PC2 and PC4 translates into yield curve gains at horizons of three to twelve months, are natural directions for future research.

Appendix

A1 Loading Stability Results

Table 10: PCA3 Reconstruction RMSE by Position

Position	Mean	Std. Dev.	Min	Q25	Median	Q75	Max
-211	0.000181	0.000191	0.000010	0.000079	0.000124	0.000213	0.002775
-190	0.000181	0.000191	0.000010	0.000079	0.000124	0.000215	0.002789
-169	0.000181	0.000191	0.000010	0.000079	0.000124	0.000214	0.002825
-148	0.000182	0.000193	0.000011	0.000079	0.000124	0.000215	0.002854
-127	0.000182	0.000195	0.000011	0.000079	0.000125	0.000216	0.002895
-106	0.000182	0.000196	0.000010	0.000079	0.000126	0.000215	0.002932
-85	0.000183	0.000197	0.000010	0.000079	0.000126	0.000216	0.002965
-64	0.000184	0.000200	0.000010	0.000079	0.000126	0.000218	0.003033
-43	0.000183	0.000192	0.000010	0.000079	0.000126	0.000218	0.002769
-22	0.000182	0.000187	0.000012	0.000079	0.000128	0.000217	0.002720
Current	0.000182	0.000188	0.000012	0.000080	0.000129	0.000219	0.002735
+21	0.000191	0.000222	0.000012	0.000081	0.000130	0.000225	0.003015

Table 11: PCA4 Reconstruction RMSE by Position

Position	Mean	Std. Dev.	Min	Q25	Median	Q75	Max
-211	0.000129	0.000120	0.000009	0.000060	0.000093	0.000155	0.001530
-190	0.000129	0.000120	0.000010	0.000060	0.000093	0.000155	0.001515
-169	0.000129	0.000121	0.000010	0.000060	0.000092	0.000153	0.001493
-148	0.000128	0.000120	0.000009	0.000059	0.000092	0.000152	0.001466
-127	0.000128	0.000119	0.000009	0.000060	0.000092	0.000151	0.001401
-106	0.000127	0.000116	0.000009	0.000060	0.000091	0.000151	0.001254
-85	0.000127	0.000115	0.000008	0.000060	0.000091	0.000151	0.001213
-64	0.000127	0.000112	0.000008	0.000059	0.000091	0.000151	0.001014
-43	0.000126	0.000108	0.000008	0.000059	0.000091	0.000151	0.001006
-22	0.000125	0.000106	0.000008	0.000060	0.000092	0.000151	0.000971
Current	0.000125	0.000106	0.000009	0.000060	0.000093	0.000152	0.001759
+21	0.000134	0.000147	0.000008	0.000061	0.000094	0.000156	0.001927

Table 12: NS Reconstruction RMSE by Position

Position	Mean	Std. Dev.	Min	Q25	Median	Q75	Max
-211	0.000625	0.000344	0.000104	0.000363	0.000539	0.000848	0.002239
-190	0.000624	0.000341	0.000105	0.000364	0.000538	0.000847	0.002219
-169	0.000621	0.000339	0.000108	0.000360	0.000536	0.000847	0.002189
-148	0.000619	0.000338	0.000112	0.000358	0.000536	0.000844	0.002153
-127	0.000617	0.000337	0.000112	0.000357	0.000535	0.000840	0.002153
-106	0.000614	0.000336	0.000113	0.000356	0.000533	0.000836	0.002153
-85	0.000612	0.000334	0.000113	0.000352	0.000531	0.000831	0.002153
-64	0.000612	0.000337	0.000113	0.000348	0.000528	0.000829	0.002153
-43	0.000616	0.000343	0.000113	0.000347	0.000528	0.000827	0.002153
-22	0.000621	0.000350	0.000113	0.000351	0.000528	0.000831	0.002319
Current	0.000630	0.000361	0.000112	0.000359	0.000530	0.000838	0.002318
+21	0.000640	0.000373	0.000112	0.000364	0.000533	0.000858	0.002317

A2 Reconstruction Error by Tenor

Table 13: PCA3 Reconstruction Errors by Tenor

Tenor	Mean	Std. Dev.	Min	Q25	Median	Q75	Max
1Y	0.000528	0.000463	0.000000	0.000059	0.000135	0.000285	0.008535
2Y	0.000220	0.000192	0.000000	0.000022	0.000050	0.000120	0.002964
3Y	0.000135	0.000103	0.000000	0.000027	0.000057	0.000107	0.000991
4Y	0.000172	0.000136	0.000000	0.000032	0.000070	0.000134	0.001918
5Y	0.000201	0.000163	0.000000	0.000032	0.000074	0.000147	0.002590
6Y	0.000200	0.000168	0.000000	0.000028	0.000063	0.000134	0.002710
7Y	0.000198	0.000171	0.000000	0.000023	0.000054	0.000113	0.002634
8Y	0.000198	0.000167	0.000000	0.000026	0.000057	0.000121	0.002237
9Y	0.000206	0.000166	0.000000	0.000031	0.000075	0.000146	0.001746
10Y	0.000208	0.000163	0.000000	0.000037	0.000083	0.000157	0.001957
12Y	0.000334	0.000261	0.000000	0.000053	0.000119	0.000255	0.002364
15Y	0.000490	0.000404	0.000000	0.000062	0.000145	0.000326	0.003834
20Y	0.000425	0.000392	0.000000	0.000040	0.000092	0.000181	0.006969
25Y	0.000241	0.000187	0.000000	0.000042	0.000094	0.000186	0.001664
30Y	0.000286	0.000226	0.000000	0.000048	0.000107	0.000209	0.002268

Table 14: PCA4 Reconstruction Errors by Tenor

Tenor	Mean	Std. Dev.	Min	Q25	Median	Q75	Max
1Y	0.000227	0.000180	0.000000	0.000033	0.000076	0.000168	0.001640
2Y	0.000075	0.000055	0.000000	0.000016	0.000034	0.000065	0.000586
3Y	0.000122	0.000095	0.000000	0.000024	0.000050	0.000093	0.001066
4Y	0.000131	0.000103	0.000000	0.000021	0.000048	0.000101	0.000997
5Y	0.000121	0.000093	0.000000	0.000021	0.000049	0.000099	0.001123
6Y	0.000097	0.000072	0.000000	0.000019	0.000041	0.000084	0.000669
7Y	0.000084	0.000060	0.000000	0.000018	0.000040	0.000078	0.000452
8Y	0.000104	0.000077	0.000000	0.000021	0.000046	0.000091	0.000784
9Y	0.000137	0.000107	0.000000	0.000024	0.000056	0.000109	0.001416
10Y	0.000157	0.000128	0.000000	0.000024	0.000054	0.000112	0.001842
12Y	0.000308	0.000254	0.000000	0.000033	0.000086	0.000209	0.002631
15Y	0.000235	0.000191	0.000000	0.000033	0.000084	0.000172	0.002804
20Y	0.000428	0.000396	0.000000	0.000039	0.000087	0.000175	0.006941
25Y	0.000187	0.000150	0.000000	0.000030	0.000068	0.000129	0.001438
30Y	0.000230	0.000185	0.000000	0.000034	0.000081	0.000165	0.002004

Table 15: NS Reconstruction Errors by Tenor

Tenor	Mean	Std. Dev.	Min	Q25	Median	Q75	Max
1Y	0.001157	0.000810	0.000000	0.000203	0.000548	0.001170	0.004470
2Y	0.000367	0.000249	0.000000	0.000089	0.000202	0.000374	0.001599
3Y	0.000401	0.000233	0.000000	0.000134	0.000294	0.000481	0.001329
4Y	0.000625	0.000392	0.000000	0.000165	0.000384	0.000700	0.001857
5Y	0.000672	0.000450	0.000000	0.000160	0.000355	0.000701	0.002220
6Y	0.000622	0.000426	0.000000	0.000159	0.000294	0.000635	0.002335
7Y	0.000516	0.000344	0.000000	0.000136	0.000295	0.000530	0.002113
8Y	0.000361	0.000225	0.000000	0.000109	0.000237	0.000395	0.001431
9Y	0.000330	0.000202	0.000000	0.000093	0.000209	0.000402	0.001022
10Y	0.000438	0.000264	0.000000	0.000143	0.000303	0.000507	0.002050
12Y	0.000842	0.000559	0.000000	0.000193	0.000483	0.000869	0.003632
15Y	0.001097	0.000744	0.000000	0.000245	0.000606	0.001083	0.004287
20Y	0.001160	0.000842	0.000000	0.000281	0.000553	0.000980	0.004816
25Y	0.000845	0.000585	0.000000	0.000200	0.000452	0.000768	0.004635
30Y	0.000840	0.000621	0.000000	0.000209	0.000462	0.000715	0.005808

A3 ACF and PACF Results

Table 16: ACF and PACF of 21-day PC Factor Changes at Selected Lags. * denotes significance at the 95% level ($|r| > 1.96/\sqrt{N}$).

Lag	PC1		PC2		PC3		PC4	
	ACF	PACF	ACF	PACF	ACF	PACF	ACF	PACF
22	-0.4888*	+0.6106*	-0.5469*	+0.4928*	-0.4420*	+0.4552*	-0.5234*	+0.5856*
43	+0.0259*	+0.4546*	+0.1131*	+0.2529*	-0.0055	+0.1902*	+0.0766*	+0.3442*
64	-0.0532*	+0.3397*	+0.0190	+0.0916*	+0.0867*	+0.0666*	-0.0741*	+0.2313*
85	+0.0994*	+0.2686*	-0.0311*	+0.0083	-0.0277*	+0.0124	+0.1467*	+0.0807*
106	-0.0318*	+0.2107*	-0.0017	+0.0046	+0.0139	-0.0072	-0.1506*	+0.0392*
127	-0.0319*	+0.1374*	+0.0275*	-0.0258*	+0.0375*	-0.0195	+0.0773*	+0.0084
148	-0.0359*	+0.1295*	-0.0759*	+0.0185	-0.1326*	+0.0129	+0.0081	+0.0130
169	+0.0437*	+0.0750*	+0.1163*	+0.0006	+0.1142*	-0.0154	-0.0471*	-0.0606*
190	+0.0551*	+0.0551*	-0.1292*	-0.0123	+0.0032	-0.0150	+0.0193	-0.0116
211	-0.0912*	+0.0299*	+0.0707*	-0.0080	-0.0806*	+0.0020	+0.0256*	+0.0156
232	+0.0176	+0.0218	+0.0295*	-0.0049	-0.0136	+0.0072	-0.0045	+0.0086
253	+0.0402*	+0.0027	-0.0434*	+0.0180	+0.0444*	+0.0065	-0.0889*	-0.0212
274	-0.0076	+0.0061	-0.0016	-0.0089	-0.0158	-0.0107	+0.1333*	-0.0339*
295	+0.0036	-0.0172	-0.0004	+0.0164	-0.0260*	+0.0035	-0.1346*	-0.0240

A4 In-Sample VAR Results

Table 17: VAR Coefficients — PC1 Equation (intercept = +0.000143, $R^2 = 0.4683$, $N = 5998$)

Lag	Predictor	Coef.	SE	CI Lo	CI Hi	t	p
22	PC1	−0.9101***	0.0136	−0.9368	−0.8834	−66.822	0.0000
	PC2	+0.0733	0.0406	−0.0062	+0.1528	1.807	0.0707
	PC3	+0.4002***	0.0822	+0.2391	+0.5614	4.869	0.0000
	PC4	+0.5955***	0.1190	+0.3622	+0.8289	5.002	0.0000
43	PC1	−0.8148***	0.0189	−0.8518	−0.7778	−43.190	0.0000
	PC2	+0.1214*	0.0506	+0.0222	+0.2205	2.400	0.0164
	PC3	+0.3974***	0.0969	+0.2074	+0.5873	4.101	0.0000
	PC4	+0.4054**	0.1534	+0.1047	+0.7061	2.643	0.0082
64	PC1	−0.7250***	0.0221	−0.7684	−0.6816	−32.734	0.0000
	PC2	−0.0325	0.0529	−0.1363	+0.0713	−0.614	0.5395
	PC3	+0.0060	0.1001	−0.1903	+0.2022	0.060	0.9525
	PC4	−0.0468	0.1690	−0.3781	+0.2845	−0.277	0.7818
85	PC1	−0.5201***	0.0238	−0.5667	−0.4735	−21.865	0.0000
	PC2	−0.0556	0.0532	−0.1598	+0.0487	−1.045	0.2962
	PC3	−0.7329***	0.1002	−0.9293	−0.5366	−7.317	0.0000
	PC4	−0.3748*	0.1755	−0.7188	−0.0308	−2.136	0.0327
106	PC1	−0.4371***	0.0241	−0.4843	−0.3899	−18.150	0.0000
	PC2	−0.3181***	0.0536	−0.4230	−0.2131	−5.938	0.0000
	PC3	−0.5556***	0.0995	−0.7506	−0.3606	−5.586	0.0000
	PC4	−0.0875	0.1766	−0.4337	+0.2586	−0.496	0.6200
127	PC1	−0.3825***	0.0239	−0.4294	−0.3355	−15.973	0.0000
	PC2	−0.3753***	0.0534	−0.4799	−0.2706	−7.030	0.0000
	PC3	−0.9185***	0.0987	−1.1120	−0.7249	−9.303	0.0000
	PC4	−0.9548***	0.1774	−1.3026	−0.6070	−5.382	0.0000
148	PC1	−0.3387***	0.0229	−0.3837	−0.2937	−14.763	0.0000
	PC2	−0.3264***	0.0533	−0.4309	−0.2219	−6.121	0.0000
	PC3	−0.6631***	0.0988	−0.8567	−0.4695	−6.714	0.0000
	PC4	−1.2480***	0.1764	−1.5938	−0.9022	−7.074	0.0000
169	PC1	−0.2070***	0.0211	−0.2485	−0.1656	−9.797	0.0000
	PC2	−0.4332***	0.0530	−0.5372	−0.3293	−8.171	0.0000
	PC3	−0.2754**	0.0984	−0.4683	−0.0825	−2.799	0.0051
	PC4	−1.2359***	0.1692	−1.5676	−0.9041	−7.303	0.0000
190	PC1	−0.0987***	0.0184	−0.1347	−0.0626	−5.362	0.0000
	PC2	−0.5259***	0.0504	−0.6248	−0.4271	−10.428	0.0000
	PC3	+0.2745**	0.0943	+0.0897	+0.4593	2.912	0.0036
	PC4	−0.9635***	0.1556	−1.2684	−0.6585	−6.193	0.0000
211	PC1	−0.0923***	0.0135	−0.1188	−0.0659	−6.841	0.0000
	PC2	−0.2098***	0.0397	−0.2877	−0.1319	−5.278	0.0000
	PC3	+0.2080*	0.0808	+0.0496	+0.3665	2.574	0.0101
	PC4	−0.2239	0.1197	−0.4585	+0.0108	−1.870	0.0616

Table 18: VAR Coefficients — PC2 Equation (intercept = +0.000143, $R^2 = 0.4668$, $N = 5998$)

Lag	Predictor	Coef.	SE	CI Lo	CI Hi	t	p
22	PC1	-0.0024	0.0045	-0.0112	+0.0065	-0.527	0.5982
	PC2	-0.6859***	0.0134	-0.7123	-0.6595	-51.003	0.0000
	PC3	-0.0182	0.0273	-0.0716	+0.0352	-0.668	0.5042
	PC4	-0.1430***	0.0395	-0.2204	-0.0656	-3.622	0.0003
43	PC1	-0.0220***	0.0063	-0.0343	-0.0098	-3.524	0.0004
	PC2	-0.3733***	0.0168	-0.4062	-0.3405	-22.270	0.0000
	PC3	-0.1913***	0.0321	-0.2543	-0.1283	-5.954	0.0000
	PC4	+0.4705***	0.0509	+0.3708	+0.5702	9.251	0.0000
64	PC1	-0.0474***	0.0073	-0.0618	-0.0330	-6.456	0.0000
	PC2	-0.1669***	0.0176	-0.2013	-0.1325	-9.507	0.0000
	PC3	-0.2035***	0.0332	-0.2686	-0.1385	-6.132	0.0000
	PC4	-0.0042	0.0560	-0.1141	+0.1056	-0.076	0.9397
85	PC1	-0.0330***	0.0079	-0.0484	-0.0175	-4.179	0.0000
	PC2	-0.1419***	0.0176	-0.1765	-0.1074	-8.047	0.0000
	PC3	+0.0391	0.0332	-0.0261	+0.1042	1.176	0.2396
	PC4	+0.0389	0.0582	-0.0751	+0.1530	0.669	0.5035
106	PC1	-0.0463***	0.0080	-0.0620	-0.0307	-5.800	0.0000
	PC2	-0.0651***	0.0178	-0.0999	-0.0303	-3.665	0.0002
	PC3	+0.1620***	0.0330	+0.0973	+0.2266	4.912	0.0000
	PC4	+0.2107***	0.0585	+0.0959	+0.3255	3.599	0.0003
127	PC1	-0.0341***	0.0079	-0.0496	-0.0185	-4.293	0.0000
	PC2	-0.0429*	0.0177	-0.0776	-0.0082	-2.426	0.0153
	PC3	+0.1059**	0.0327	+0.0417	+0.1701	3.235	0.0012
	PC4	+0.2690***	0.0588	+0.1537	+0.3843	4.573	0.0000
148	PC1	-0.0192*	0.0076	-0.0341	-0.0043	-2.525	0.0116
	PC2	-0.0913***	0.0177	-0.1259	-0.0566	-5.161	0.0000
	PC3	+0.0741*	0.0327	+0.0099	+0.1383	2.263	0.0237
	PC4	+0.2168***	0.0585	+0.1022	+0.3315	3.707	0.0002
169	PC1	-0.0272***	0.0070	-0.0409	-0.0135	-3.883	0.0001
	PC2	-0.0165	0.0176	-0.0510	+0.0180	-0.938	0.3482
	PC3	+0.0813*	0.0326	+0.0174	+0.1453	2.493	0.0127
	PC4	+0.5356***	0.0561	+0.4256	+0.6456	9.546	0.0000
190	PC1	-0.0278***	0.0061	-0.0398	-0.0158	-4.557	0.0000
	PC2	-0.0675***	0.0167	-0.1003	-0.0347	-4.037	0.0001
	PC3	+0.1058***	0.0313	+0.0445	+0.1671	3.384	0.0007
	PC4	+0.1752***	0.0516	+0.0741	+0.2763	3.396	0.0007
211	PC1	-0.0069	0.0045	-0.0157	+0.0018	-1.550	0.1212
	PC2	-0.0242	0.0132	-0.0500	+0.0016	-1.836	0.0664
	PC3	-0.1025***	0.0268	-0.1551	-0.0500	-3.826	0.0001
	PC4	+0.2660***	0.0397	+0.1882	+0.3438	6.701	0.0000

Table 19: VAR Coefficients — PC3 Equation (intercept = -0.000059 , $R^2 = 0.3362$, $N = 5998$)

Lag	Predictor	Coef.	SE	CI Lo	CI Hi	t	p
22	PC1	+0.0150***	0.0022	+0.0106	+0.0194	6.672	0.0000
	PC2	+0.0123	0.0067	-0.0009	+0.0254	1.833	0.0669
	PC3	-0.5781***	0.0135	-0.6047	-0.5515	-42.668	0.0000
	PC4	-0.0227	0.0196	-0.0611	+0.0158	-1.154	0.2484
43	PC1	+0.0325***	0.0031	+0.0264	+0.0386	10.453	0.0000
	PC2	+0.0046	0.0083	-0.0118	+0.0209	0.547	0.5845
	PC3	-0.2933***	0.0160	-0.3247	-0.2620	-18.367	0.0000
	PC4	-0.0652*	0.0253	-0.1147	-0.0156	-2.577	0.0100
64	PC1	+0.0259***	0.0037	+0.0187	+0.0330	7.082	0.0000
	PC2	+0.0179*	0.0087	+0.0008	+0.0350	2.047	0.0407
	PC3	-0.0777***	0.0165	-0.1101	-0.0454	-4.710	0.0000
	PC4	-0.1037***	0.0279	-0.1583	-0.0491	-3.723	0.0002
85	PC1	+0.0408***	0.0039	+0.0331	+0.0485	10.414	0.0000
	PC2	+0.0320***	0.0088	+0.0148	+0.0492	3.646	0.0003
	PC3	+0.0263	0.0165	-0.0061	+0.0586	1.590	0.1119
	PC4	-0.0676*	0.0289	-0.1243	-0.0109	-2.335	0.0196
106	PC1	+0.0312***	0.0040	+0.0234	+0.0390	7.861	0.0000
	PC2	-0.0025	0.0088	-0.0198	+0.0149	-0.278	0.7810
	PC3	+0.0937***	0.0164	+0.0616	+0.1259	5.717	0.0000
	PC4	+0.2235***	0.0291	+0.1664	+0.2805	7.678	0.0000
127	PC1	+0.0064	0.0039	-0.0013	+0.0142	1.632	0.1027
	PC2	+0.0024	0.0088	-0.0149	+0.0196	0.271	0.7862
	PC3	+0.0662***	0.0163	+0.0343	+0.0981	4.067	0.0000
	PC4	+0.2066***	0.0292	+0.1492	+0.2639	7.063	0.0000
148	PC1	+0.0135***	0.0038	+0.0060	+0.0209	3.559	0.0004
	PC2	+0.0145	0.0088	-0.0028	+0.0317	1.644	0.1002
	PC3	-0.0664***	0.0163	-0.0983	-0.0345	-4.077	0.0000
	PC4	+0.1030***	0.0291	+0.0460	+0.1600	3.542	0.0004
169	PC1	+0.0144***	0.0035	+0.0075	+0.0212	4.126	0.0000
	PC2	-0.0273**	0.0087	-0.0444	-0.0102	-3.123	0.0018
	PC3	+0.0140	0.0162	-0.0178	+0.0458	0.863	0.3882
	PC4	+0.0658*	0.0279	+0.0111	+0.1205	2.358	0.0184
190	PC1	+0.0083**	0.0030	+0.0024	+0.0143	2.741	0.0061
	PC2	-0.0190*	0.0083	-0.0353	-0.0027	-2.285	0.0223
	PC3	+0.0137	0.0155	-0.0168	+0.0442	0.881	0.3786
	PC4	-0.0533*	0.0256	-0.1036	-0.0030	-2.079	0.0377
211	PC1	+0.0031	0.0022	-0.0013	+0.0074	1.379	0.1680
	PC2	+0.0024	0.0066	-0.0105	+0.0152	0.362	0.7176
	PC3	-0.0363**	0.0133	-0.0624	-0.0102	-2.726	0.0064
	PC4	+0.0085	0.0197	-0.0302	+0.0471	0.428	0.6685

Table 20: VAR Coefficients — PC4 Equation (intercept = -0.000000 , $R^2 = 0.4581$, $N = 5998$)

Lag	Predictor	Coef.	SE	CI Lo	CI Hi	t	p
22	PC1	+0.0102***	0.0016	+0.0072	+0.0133	6.524	0.0000
	PC2	-0.0195***	0.0047	-0.0286	-0.0103	-4.170	0.0000
	PC3	+0.0090	0.0095	-0.0095	+0.0276	0.954	0.3401
	PC4	-0.7699***	0.0137	-0.7967	-0.7430	-56.179	0.0000
43	PC1	+0.0168***	0.0022	+0.0126	+0.0211	7.742	0.0000
	PC2	-0.0136*	0.0058	-0.0250	-0.0022	-2.333	0.0197
	PC3	+0.0078	0.0112	-0.0141	+0.0296	0.695	0.4871
	PC4	-0.5772***	0.0177	-0.6118	-0.5426	-32.692	0.0000
64	PC1	+0.0182***	0.0025	+0.0132	+0.0232	7.142	0.0000
	PC2	-0.0128*	0.0061	-0.0247	-0.0008	-2.099	0.0359
	PC3	+0.0773***	0.0115	+0.0547	+0.0999	6.711	0.0000
	PC4	-0.4002***	0.0195	-0.4384	-0.3621	-20.574	0.0000
85	PC1	+0.0153***	0.0027	+0.0099	+0.0207	5.586	0.0000
	PC2	-0.0376***	0.0061	-0.0496	-0.0256	-6.144	0.0000
	PC3	-0.0384***	0.0115	-0.0610	-0.0158	-3.330	0.0009
	PC4	-0.2254***	0.0202	-0.2650	-0.1858	-11.161	0.0000
106	PC1	+0.0116***	0.0028	+0.0062	+0.0170	4.186	0.0000
	PC2	+0.0060	0.0062	-0.0061	+0.0181	0.976	0.3291
	PC3	-0.0202	0.0114	-0.0426	+0.0023	-1.761	0.0783
	PC4	-0.1961***	0.0203	-0.2359	-0.1562	-9.648	0.0000
127	PC1	+0.0120***	0.0028	+0.0066	+0.0174	4.342	0.0000
	PC2	-0.0396***	0.0061	-0.0516	-0.0276	-6.445	0.0000
	PC3	-0.0554***	0.0114	-0.0777	-0.0331	-4.874	0.0000
	PC4	-0.1328***	0.0204	-0.1728	-0.0928	-6.503	0.0000
148	PC1	+0.0154***	0.0026	+0.0102	+0.0206	5.829	0.0000
	PC2	-0.0408***	0.0061	-0.0528	-0.0288	-6.645	0.0000
	PC3	-0.0001	0.0114	-0.0224	+0.0221	-0.013	0.9896
	PC4	-0.1198***	0.0203	-0.1596	-0.0800	-5.899	0.0000
169	PC1	+0.0173***	0.0024	+0.0125	+0.0221	7.113	0.0000
	PC2	-0.0133*	0.0061	-0.0253	-0.0014	-2.184	0.0290
	PC3	-0.0464***	0.0113	-0.0686	-0.0242	-4.095	0.0000
	PC4	-0.1029***	0.0195	-0.1411	-0.0648	-5.284	0.0000
190	PC1	+0.0120***	0.0021	+0.0079	+0.0162	5.668	0.0000
	PC2	+0.0123*	0.0058	+0.0010	+0.0237	2.127	0.0335
	PC3	+0.0373***	0.0109	+0.0160	+0.0585	3.433	0.0006
	PC4	+0.0126	0.0179	-0.0225	+0.0477	0.702	0.4829
211	PC1	+0.0006	0.0016	-0.0025	+0.0036	0.360	0.7185
	PC2	-0.0042	0.0046	-0.0132	+0.0048	-0.922	0.3567
	PC3	+0.0213*	0.0093	+0.0031	+0.0395	2.289	0.0221
	PC4	+0.0260	0.0138	-0.0010	+0.0531	1.889	0.0589

A5 Out-of-Sample LSTM Forecast Results

Table 21: OOS rolling LSTM performance — LSTM-VAR.

PC	N	Accuracy	R^2	RMSE	AIC	BIC
PC1	2977	64.70%	0.0949	0.010304	-18710.59	-18464.64
PC2	2977	56.73%	-0.2481	0.003134	-25796.53	-25550.58
PC3	2977	59.05%	-0.1547	0.001291	-31075.71	-30829.77
PC4	2977	57.10%	-0.6572	0.000665	-35030.10	-34784.16

Table 22: OOS forecast residual statistics — LSTM-VAR.

PC	Mean	Std. Dev.	Min	Q25	Q75	Max
PC1	0.000495	0.010292	-0.047743	-0.005222	0.006387	0.053006
PC2	0.000019	0.003134	-0.012220	-0.001668	0.001606	0.015873
PC3	0.000073	0.001289	-0.004903	-0.000682	0.000792	0.009416
PC4	-0.000018	0.000664	-0.002706	-0.000385	0.000314	0.005576

Table 23: OOS rolling LSTM performance — LSTM-FULL.

PC	N	Accuracy	R^2	RMSE	AIC	BIC
PC1	2977	70.27%	0.2355	0.009470	-19213.25	-18967.30
PC2	2977	68.26%	0.1577	0.002575	-26967.33	-26721.39
PC3	2977	65.97%	0.0955	0.001143	-31802.90	-31556.95
PC4	2977	65.50%	-0.1289	0.000549	-36172.71	-35926.77

Table 24: OOS forecast residual statistics — LSTM-FULL.

PC	Mean	Std. Dev.	Min	Q25	Q75	Max
PC1	0.000686	0.009445	−0.042103	−0.004543	0.005792	0.049596
PC2	−0.000190	0.002568	−0.011084	−0.001502	0.001169	0.010648
PC3	0.000047	0.001142	−0.004319	−0.000612	0.000653	0.006796
PC4	−0.000003	0.000549	−0.002388	−0.000280	0.000269	0.003276

Table 25: OOS rolling LSTM performance — LSTM-FULL-EIG.

PC	N	Accuracy	R^2	RMSE	AIC	BIC
PC1	2977	69.70%	0.2569	0.009336	−19297.61	−19051.67
PC2	2977	65.57%	0.1374	0.002606	−26896.28	−26650.34
PC3	2977	65.77%	0.1759	0.001091	−32079.85	−31833.91
PC4	2977	55.59%	−0.1117	0.000544	−36218.65	−35972.71

Table 26: OOS forecast residual statistics — LSTM-FULL-EIG.

PC	Mean	Std. Dev.	Min	Q25	Q75	Max
PC1	0.000552	0.009320	−0.042103	−0.004527	0.005034	0.051141
PC2	−0.000048	0.002605	−0.015010	−0.001317	0.001230	0.014038
PC3	0.000011	0.001091	−0.004824	−0.000599	0.000581	0.006190
PC4	0.000027	0.000544	−0.002232	−0.000255	0.000296	0.003725

A6 Yield Curve Reconstruction Residual Statistics

Table 27: Yield curve forecast residual statistics by tenor — VAR.

Tenor	Mean	Std. Dev.	Min	Q25	Q75	Max
1Y	−0.000251	0.002186	−0.015130	−0.000822	0.000704	0.008654
2Y	−0.000231	0.002264	−0.015098	−0.000871	0.000723	0.011291
3Y	−0.000215	0.002365	−0.015203	−0.001001	0.000838	0.012734
4Y	−0.000202	0.002457	−0.015302	−0.001161	0.000967	0.013394
5Y	−0.000189	0.002532	−0.015514	−0.001285	0.001079	0.013530
6Y	−0.000178	0.002587	−0.015634	−0.001374	0.001175	0.013258
7Y	−0.000166	0.002636	−0.015604	−0.001467	0.001251	0.012758
8Y	−0.000155	0.002689	−0.015543	−0.001509	0.001348	0.012215
9Y	−0.000146	0.002743	−0.015413	−0.001556	0.001413	0.011616
10Y	−0.000141	0.002788	−0.015480	−0.001586	0.001446	0.011310
12Y	−0.000137	0.002901	−0.015846	−0.001635	0.001526	0.010757
15Y	−0.000112	0.002956	−0.014856	−0.001693	0.001626	0.010951
20Y	−0.000097	0.003079	−0.015306	−0.001781	0.001716	0.011352
25Y	−0.000083	0.003065	−0.013591	−0.001876	0.001773	0.011474
30Y	−0.000085	0.003049	−0.012177	−0.001869	0.001742	0.011737

Table 28: Yield curve forecast residual statistics by tenor — LSTM-VAR-FULL-EIG-IV.

Tenor	Mean	Std. Dev.	Min	Q25	Q75	Max
1Y	-0.000261	0.002494	-0.019234	-0.000948	0.000771	0.008934
2Y	-0.000291	0.002591	-0.020290	-0.001057	0.000812	0.010103
3Y	-0.000311	0.002698	-0.020644	-0.001251	0.000909	0.011281
4Y	-0.000324	0.002790	-0.020597	-0.001421	0.001013	0.012129
5Y	-0.000329	0.002860	-0.020328	-0.001528	0.001081	0.012518
6Y	-0.000329	0.002903	-0.019772	-0.001622	0.001164	0.012463
7Y	-0.000325	0.002934	-0.018957	-0.001689	0.001250	0.012663
8Y	-0.000319	0.002969	-0.018091	-0.001781	0.001351	0.012837
9Y	-0.000314	0.003005	-0.017306	-0.001895	0.001419	0.013011
10Y	-0.000311	0.003037	-0.017154	-0.001916	0.001459	0.013123
12Y	-0.000310	0.003106	-0.017095	-0.001999	0.001482	0.013353
15Y	-0.000288	0.003175	-0.016667	-0.001996	0.001597	0.013325
20Y	-0.000282	0.003270	-0.017451	-0.002122	0.001669	0.013163
25Y	-0.000277	0.003248	-0.016018	-0.002174	0.001719	0.013066
30Y	-0.000278	0.003234	-0.014674	-0.002182	0.001710	0.013089

Table 29: Yield curve forecast residual statistics by tenor — Hybrid (VAR PC1, PC3; LSTM PC2, PC4).

Tenor	Mean	Std. Dev.	Min	Q25	Q75	Max
1Y	-0.000214	0.002274	-0.016068	-0.000886	0.000735	0.008371
2Y	-0.000208	0.002382	-0.016187	-0.000976	0.000749	0.010633
3Y	-0.000204	0.002487	-0.016375	-0.001102	0.000864	0.012245
4Y	-0.000198	0.002570	-0.016380	-0.001231	0.001022	0.012954
5Y	-0.000191	0.002627	-0.016310	-0.001370	0.001127	0.013058
6Y	-0.000183	0.002658	-0.016152	-0.001429	0.001187	0.012866
7Y	-0.000173	0.002680	-0.015885	-0.001530	0.001268	0.012449
8Y	-0.000163	0.002707	-0.015578	-0.001556	0.001361	0.012075
9Y	-0.000155	0.002738	-0.015208	-0.001560	0.001406	0.011736
10Y	-0.000149	0.002765	-0.015065	-0.001596	0.001441	0.011589
12Y	-0.000142	0.002842	-0.015003	-0.001659	0.001517	0.011159
15Y	-0.000124	0.002902	-0.014651	-0.001679	0.001609	0.011627
20Y	-0.000121	0.002997	-0.014895	-0.001759	0.001645	0.011612
25Y	-0.000112	0.002971	-0.013561	-0.001849	0.001666	0.011194
30Y	-0.000114	0.002955	-0.012511	-0.001859	0.001656	0.010918

Table 30: Yield curve forecast residual statistics by tenor — Random walk benchmark.

Tenor	Mean	Std. Dev.	Min	Q25	Q75	Max
1Y	−0.000171	0.001696	−0.010834	−0.000539	0.000531	0.006841
2Y	−0.000173	0.001765	−0.010308	−0.000597	0.000591	0.007507
3Y	−0.000173	0.001836	−0.009891	−0.000720	0.000679	0.007764
4Y	−0.000170	0.001887	−0.009708	−0.000880	0.000778	0.007921
5Y	−0.000165	0.001920	−0.009445	−0.001002	0.000862	0.007741
6Y	−0.000159	0.001943	−0.009059	−0.001082	0.000932	0.007739
7Y	−0.000152	0.001963	−0.008708	−0.001154	0.000997	0.007696
8Y	−0.000145	0.001987	−0.008535	−0.001202	0.001060	0.007692
9Y	−0.000138	0.002011	−0.008405	−0.001220	0.001105	0.007700
10Y	−0.000131	0.002026	−0.008263	−0.001228	0.001141	0.007698
12Y	−0.000120	0.002087	−0.008002	−0.001258	0.001196	0.007845
15Y	−0.000119	0.002132	−0.008155	−0.001291	0.001251	0.007082
20Y	−0.000109	0.002194	−0.008748	−0.001308	0.001248	0.007459
25Y	−0.000103	0.002175	−0.008421	−0.001340	0.001254	0.006821
30Y	−0.000105	0.002176	−0.008609	−0.001344	0.001253	0.006625

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